

Tramacastilla_disturbance_palaeoecology

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This is the code used to obtain core data to perform statistical analyses included in file “tramacastilla_disturbance_palaeoecology_analyses” from Tramacastilla Lake sequence, included in Julián-Posada et al. (in prep).

1. Load ages, depths and sedaDNA data

```
head(tram20_depth_ages) # ages and depths for core TRAM20-1B
```

```
##           SampleId composite_depth depth min  max median mean
## 1 TRAM20-1B-3U-19-20           390 390.9 772  937   863  861
## 2 TRAM20-1B-3U-20-21           392 392.9 803 1016   902  903
## 3 TRAM20-1B-3U-21-22           394 394.9 814 1104   939  945
## 4 TRAM20-1B-3U-22-23           395 395.9 820 1149   959  967
## 5 TRAM20-1B-3U-23-24           397 397.9 834 1238   999 1009
## 6 TRAM20-1B-3U-24-25           398 398.9 838 1282  1020 1030
```

```
tram20_depth_ages_sed_rate <- tram20_depth_ages %>% # perform sedimentation rate
  rename("age" = "median") %>%
  dplyr::select(-depth) %>%
  rename("depth" = "composite_depth") %>%
  mutate(depth_diff = c(NA, diff(depth)),
         age_diff = c(NA, diff(age)),
         sed_rate = depth_diff/age_diff)
```

```
head(RAI_plants) #sedaDNA plant data
```

```
## # A tibble: 6 x 426
##   sample_id      depth  age 'Abies alba' 'Acer 1' 'Acer 2' 'Acinos arvensis'
##   <chr>          <dbl> <dbl>      <dbl>    <dbl>    <dbl>      <dbl>
## 1 TRAM_1b_3U_11.12  366 1331      0.0142  0         0         0
## 2 TRAM_1b_3U_16.17  373 1496      0.0909  0.000381  0.000101  0
## 3 TRAM_1b_3U_21.22  380 1635      0.0389  0         0         0
## 4 TRAM_1b_3U_26.27  387 1776      0.0334  0         0         0.00000306
## 5 TRAM_1b_3U_31.32  394 1920      0.0721  0         0         0
## 6 TRAM_1b_3U_36.37  401 2042      0.0634  0         0         0.0000109
## # i 419 more variables: 'Actaea spicata' <dbl>, Agrostidinae <dbl>,
## #   Ajuga <dbl>, 'Ajuga chamaepitys' <dbl>, Alchemilla <dbl>, Allium <dbl>,
## #   Alnus <dbl>, 'Alnus alnobetula' <dbl>, Narcissus <dbl>,
```

```
## # Amblystegiaceae <dbl>, Anacardiaceae <dbl>, 'Andreaea alpestris' <dbl>,
## # 'Anthyllis vulneraria' <dbl>, 'apioid superclade' <dbl>,
## # 'Apiioideae 1' <dbl>, 'Apiioideae 2' <dbl>, 'Apiioideae 3' <dbl>,
## # 'Aquilegia vulgaris' <dbl>, Arabis <dbl>, Arachis <dbl>, ...
```

```
head(RAI_animals) #sedaDNA animal data
```

```
## # A tibble: 6 x 32
##   SampleId age 'Bos taurus' 'Ovis aries' 'Cervus elaphus' 'Bufo bufo/spinosus'
##   <chr>     <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
## 1 TRAM_1b~ 1331         0.819         0.167         0             0
## 2 TRAM_1b~ 1496         0             0             0             0
## 3 TRAM_1b~ 1635         0.132         0             0             0
## 4 TRAM_1b~ 1776         0.502         0.461         0             0
## 5 TRAM_1b~ 1920         0.694         0.268         0             0
## 6 TRAM_1b~ 2042         0.408         0.575         0             0
## # i 26 more variables: 'Bos sp.' <dbl>, 'Ursus arctos' <dbl>,
## # 'Capra hircus' <dbl>, 'Rana temporaria' <dbl>, 'Sus scrofa' <dbl>,
## # 'Lissotriton helveticus' <dbl>, 'Canis lupus' <dbl>,
## # 'Pelophylax perezi' <dbl>, 'Mareca strepera/penelope' <dbl>,
## # 'Apodemus flavicollis' <dbl>, 'Alytes obstetricans' <dbl>,
## # 'Neomys fodiens' <dbl>, 'Anas platyrhynchos' <dbl>,
## # 'Psophus stridulus' <dbl>, 'Equus caballus' <dbl>, ...
```

```
RAI_animals_domestic <- RAI_animals %>% #only data from domestic animals
  rename("Cattle" = `Bos taurus`,
         "Sheep" = `Ovis aries`,
         "Goat" = `Capra hircus`) %>%
  dplyr::select (age, Cattle, Sheep, Goat)
```

Apply binning approach to sedaDNA data in order to compare it with pollen and charcoal data

```
bins_pollen <- tram20_pollen_holocene$age

RAI_animals_domestic_binned <- RAI_animals_domestic %>%
  mutate(binned_age = sapply(age, function(x) bins_pollen[which.min(abs(bins_pollen-x))])) %>%
  group_by(binned_age) %>%
  summarise(Cattle = mean(Cattle, na.rm = TRUE),
            Sheep = mean(Sheep, na.rm = TRUE),
            Goat = mean(Goat, na.rm = TRUE)) %>%
  rename("age" = "binned_age") %>%
  ungroup() %>%
  as.data.frame()
```

2. Tramacastilla charcoal data

2.1. Data upload

```
head(charcoal.tram) #charcoal data from Tramacastilla (core TRAM20-1B) obtained using Win seedle Pro 20.
```

```
##      SampleId ImageId ImageType ObjectId      DPI.ProjAreaObj
## 1 TRAM-3U-19-20      1    GLOBAL  Scanner 0.0000 35.211270 34.843210
## 2 TRAM-3U-19-20      1    SEEDLE      1      18403.1051
## 3 TRAM-3U-19-20      2    GLOBAL  Scanner 0.0000 35.211270 34.843210
## 4 TRAM-3U-20-21      1    GLOBAL  Scanner 0.0000 35.211270 34.843210
## 5 TRAM-3U-20-21      1    SEEDLE      1      7361.242
## 6 TRAM-3U-20-21      2    GLOBAL  Scanner 0.0000 35.211270 34.843210
##      NPath.WLRatio TotalProjAreaObj.Hpos NofSeedles
## 1      0.0000      18403.105      1
## 2      0.2858      21110.330     NA
## 3      0.0000      0.000      0
## 4      0.0000      7361.242      1
## 5      1.0063      20897.889     NA
## 6      0.0000      12268.737      1
```

```
head(samples_names) #names of charcoal samples
```

```
##      SampleId
## 1 TRAM-3U-19-20
## 2 TRAM-3U-20-21
## 3 TRAM-3U-21-22
## 4 TRAM-3U-22-23
## 5 TRAM-3U-23-24
## 6 TRAM-3U-24-25
```

```
Tramacastilla_age <- tram20_depth_ages %>% #ages and depths for charcoal samples
  select(SampleId, median, composite_depth) %>%
  rename("age" = "median",
         "depth" = "composite_depth") %>%
  mutate(age = as.numeric(age),
         depth = as.numeric(depth)) %>%
  mutate(SampleId = str_replace_all(SampleId, "TRAM20-1B-", "TRAM-")) %>%
  filter(SampleId %in% samples_names$SampleId) %>%
  mutate_all(~replace(., is.na(.), 0))
```

There are 194 instead of 202 samples, as some of them have no data (no charcoal particles were found), so add them.

```
charcoal.tram.202 <- full_join(charcoal.tram, samples_names, by = "SampleId") %>%
  mutate(ImageType = ifelse(is.na(ImageType), "GLOBAL", ImageType)) #change ImageType to say GLOBAL for
length(unique(charcoal.tram.202$SampleId)) #make sure there are 202 samples
```

```
## [1] 202
```

Now duplicate these new rows to obtain one row for GLOBAL and other for SEEDLE

```

samples_no_data <- charcoal.tram.202 %>%
  filter(is.na(ImageId)) %>%
  mutate(ImageType = "SEEDLE")

charcoal.tram.all <- rbind(charcoal.tram.202, samples_no_data) %>%
  mutate_all(~replace(., is.na(.), 0))

length(unique(charcoal.tram.all$SampleId)) #make sure there are 202 samples

```

```
## [1] 202
```

Create 2 dataframes: one for global and another for seedle

```

Gchar <- subset(charcoal.tram.all, ImageType=="GLOBAL")
Schar <- charcoal.tram.all %>%
  subset(ImageType=="SEEDLE") %>%
  mutate(NPath.WLRatio = ifelse(NPath.WLRatio > 1, 1/NPath.WLRatio, NPath.WLRatio)) %>% #change values
  mutate(NofSeedles = as.numeric(NofSeedles)) %>%
  mutate(DPI.ProjAreaObj = as.numeric(DPI.ProjAreaObj)) %>%
  mutate(ObjectId = as.numeric(ObjectId))

length(unique(Gchar$SampleId)) #make sure there are 202 samples for GLOBAL

```

```
## [1] 202
```

```
length(unique(Schar$SampleId)) #make sure there are 202 samples for SEEDLE
```

```
## [1] 202
```

2.2. Make a new data frame only with charcoal particles larger than 22500 μm

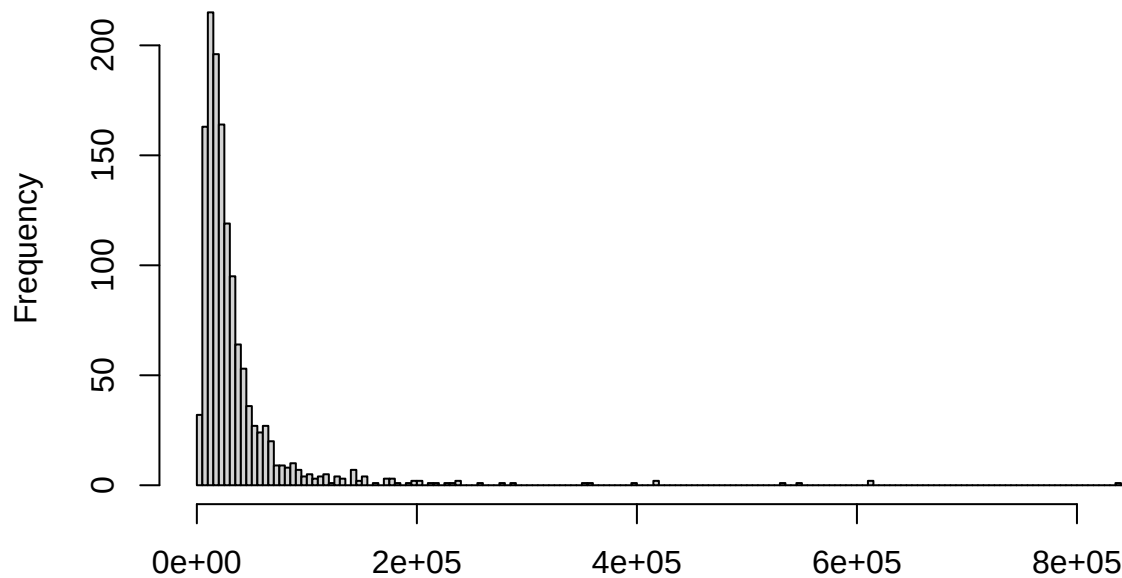
First see the frequencies of area

```

hist(Schar$DPI.ProjAreaObj, breaks = seq(0, 850000, 5000),
     xlab = "Frequency of areas of charcoal particles",
     main = paste("Area of charcoal particles"))

```

Area of charcoal particles



Frequency of areas of charcoal particles

```
summary(Schar$DPI.ProjAreaObj)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	0	13496	22084	35713	36806	836728	2

25% of the particles are below 13496 μm^2 , which is a sieve diameter of 116 μm , and mean 35713. Now filter particles so remove all that are below 110 μm , that we assume is the noisy part of the signal $\rightarrow$ remove those charcoal with an area $< 110 \mu\text{m}$

```
seedles <- charcoal.tram.all %>%
  filter(ImageType == "SEEDLE" & DPI.ProjAreaObj > 12100) %>%
  mutate(NPath.WLRatio = ifelse(NPath.WLRatio > 1, 1/NPath.WLRatio, NPath.WLRatio)) %>% #change values
  as.data.frame() %>%
  mutate(DPI.ProjAreaObj = as.numeric(DPI.ProjAreaObj))
```

Now create another data frame with two columns, SampleId and CharCount, that is the number of particles that meet the requirements

```
counting <- seedles %>%
  group_by(SampleId) %>%
  summarise(CharCount = n())
```

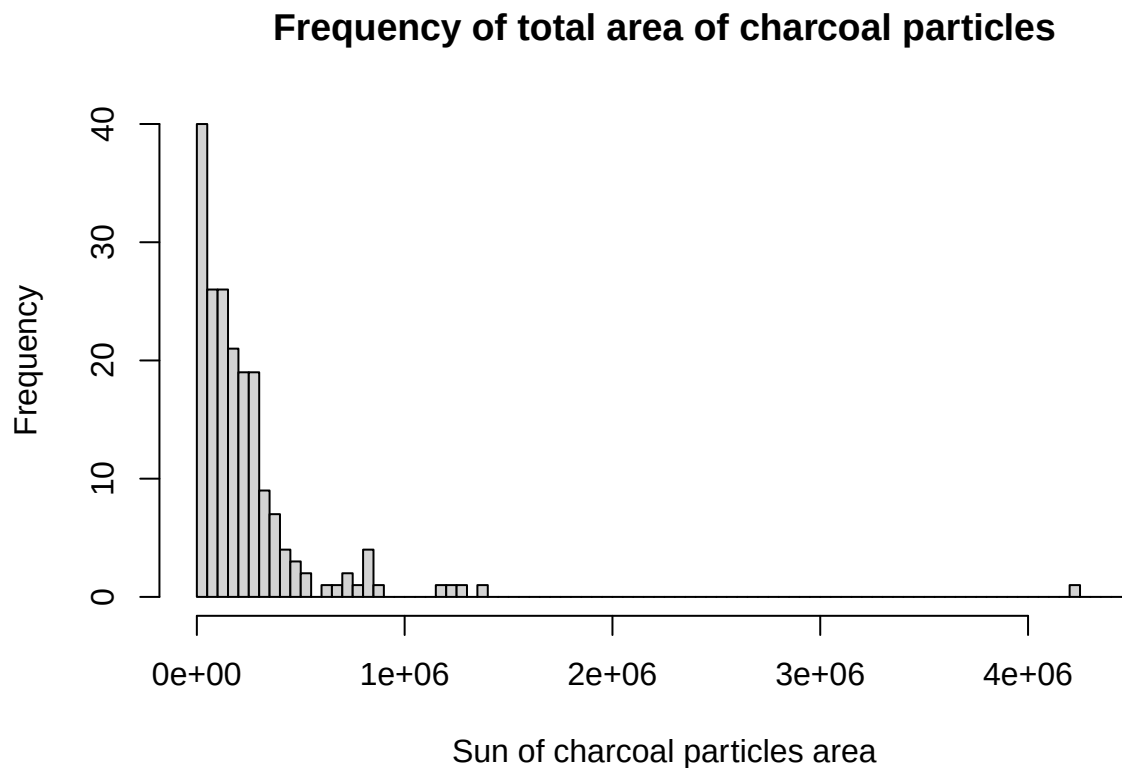
Another data frame with two columns, SampleId and sumCharArea, that is the sum of area of particles that meet the requirements

```
totalarea <- seedles %>%
  group_by(SampleId) %>%
  summarise(sumCharArea = sum(DPI.ProjAreaObj, na.rm = TRUE))

summary(totalarea$sumCharArea)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      8588  62571 160721  240249  275433 4230260
```

```
hist(totalarea$sumCharArea, breaks = seq(0, 4500000, 50000),
     main = paste("Frequency of total area of charcoal particles"),
     xlab = "Sun of charcoal particles area")
```



```
seedles.age <- seedles %>%
  merge(Tramacastilla_age, by = "SampleId") %>%
  group_by(SampleId) %>%
  mutate(Color = ifelse(median(NPath.WLRatio) > 0.5, "green", "yellow")) %>%
  distinct() %>%
  ungroup() %>%
  as.data.frame()
```

2.2.1. Create a data frame for plotting

Create a file with one value per sample: count=length sampleID, W:L= mean W:L, area= sum ProjArea

```
CharTotS_G.110 <- merge(counting, totalarea)
CharTotS_G.110.age <- merge(CharTotS_G.110, Tramacastilla_age, by = "SampleId")
```

Check if area and charcoal number are correlated

```
cor.test(CharTotS_G.110.age$sumCharArea, CharTotS_G.110.age$CharCount, method="spearman", exact = FALSE)
```

```
##
## Spearman's rank correlation rho
##
## data: CharTotS_G.110.age$sumCharArea and CharTotS_G.110.age$CharCount
## S = 159465, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.8626815
```

87% of the data is correlated. Now check if that correlation is linear or not

```
summary(lm(CharTotS_G.110.age$sumCharArea~CharTotS_G.110.age$CharCount))
```

```
##
## Call:
## lm(formula = CharTotS_G.110.age$sumCharArea ~ CharTotS_G.110.age$CharCount)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -424925  -85146   -5418   38736 1024557
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -59588     17264  -3.452 0.000688 ***
## CharTotS_G.110.age$CharCount    44776      1733  25.835 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 176600 on 189 degrees of freedom
## Multiple R-squared:  0.7793, Adjusted R-squared:  0.7781
## F-statistic: 667.4 on 1 and 189 DF, p-value: < 2.2e-16
```

78% of data is linearly correlated Join these two data frames to have all good charcoal particles with CharCount column

```
CharTotS_S.110 <- seedles %>%
  group_by(SampleId) %>%
  summarise(meanCharWL = mean(NPath.WLRatio, na.rm = TRUE),
            medianCharWL = median(NPath.WLRatio, na.rm = TRUE))

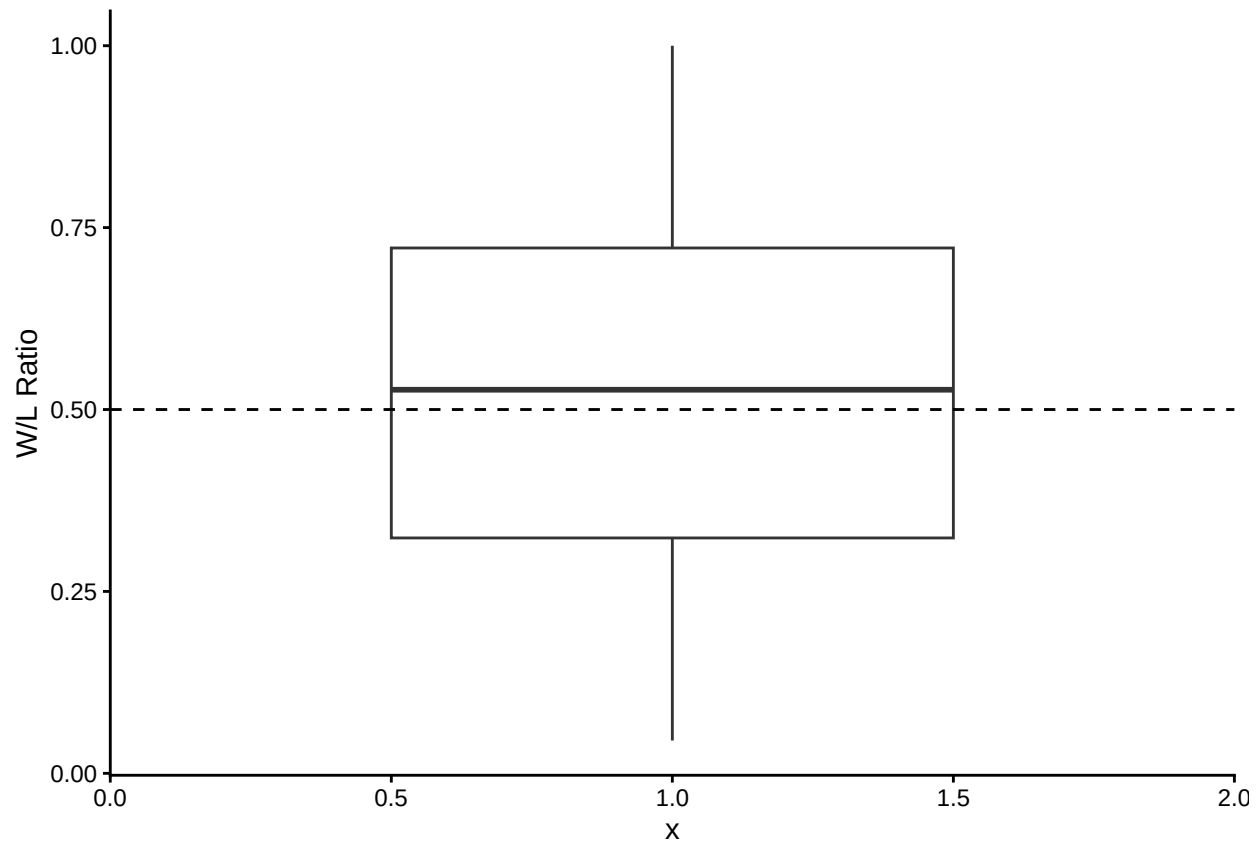
CharTotS.110 <- CharTotS_G.110 %>%
  full_join(CharTotS_S.110, by = "SampleId") %>%
  full_join(samples_names, by = "SampleId") %>% #add samples that are missing
  mutate_all(~replace(., is.na(.), 0))
```

Add ages to order the samples

```
CharTotS.110.age <- full_join(CharTotS.110, Tramacastilla_age, by = "SampleId")
```

2.2.2. Graphic representation

```
seedles %>% #min, max, mean of WLRatio
  ggplot(aes(x= 1, y = NPath.WLRatio)) +
  geom_boxplot(outlier.shape = 20, width = 1) +
  geom_hline(yintercept = 0.5, linetype = "dashed") +
  scale_x_continuous(limits = c(0, 2), expand = c(0, 0)) +
  theme_classic() +
  ylab("W/L Ratio")
```

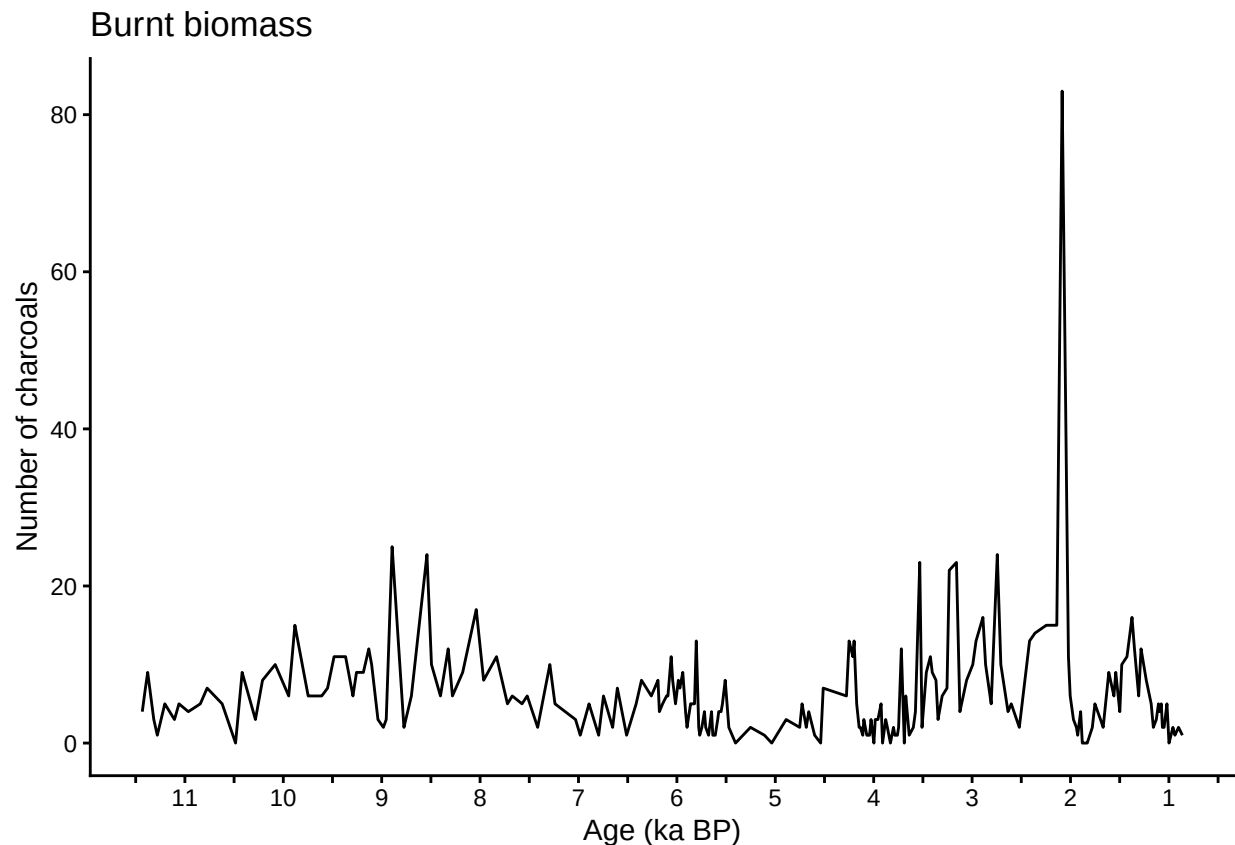


2.2.3. CharCount

```
CharTotS.110.age %>%
  ggplot(aes(x = age/1000, y = CharCount)) +
  geom_line() +
  xlab("Age (ka BP)") +
  ylab("Number of charcoals") +
```



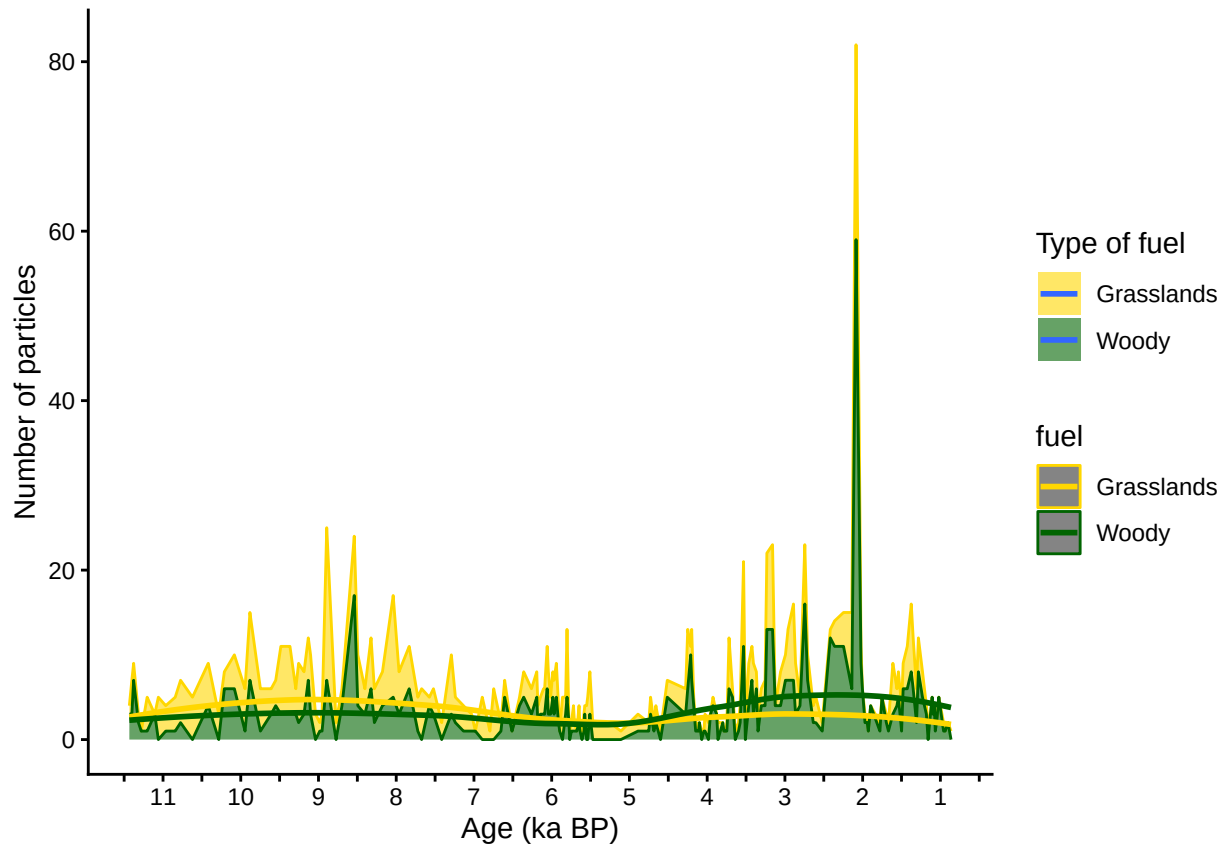
```
ggtitle("Burnt biomass") +
theme_classic() +
scale_x_reverse(breaks = seq(0.5, 12, by = 0.5),
               labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5))
```



2.2.4. W/L Ratio

```
W.L.ratio <- seedles.age %>%
  group_by(age, depth) %>%
  summarise(Woody = sum(NPath.WLRatio > 0.5),
            Grasslands = sum(NPath.WLRatio < 0.5)) %>%
  pivot_longer(cols = c("Woody", "Grasslands"), names_to = "fuel", values_to = "count")

W.L.ratio %>%
  ggplot(aes(x = age/1000, y = count, fill = fuel, color = fuel)) +
  geom_area(alpha = 0.6) +
  geom_smooth(method = "loess", se = FALSE) +
  labs(x = "Age (ka BP)", y = "Number of particles", fill = "Type of fuel") +
  theme_classic() +
  scale_x_reverse(limits = c(max(W.L.ratio$age)/1000, min(W.L.ratio$age)/1000),
                 breaks = seq(0.5, 12, by = 0.5),
                 labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5))
  scale_fill_manual(values = c("Woody" = "darkgreen", "Grasslands" = "#FFD700")) +
  scale_color_manual(values = c("Woody" = "darkgreen", "Grasslands" = "#FFD700"))
```

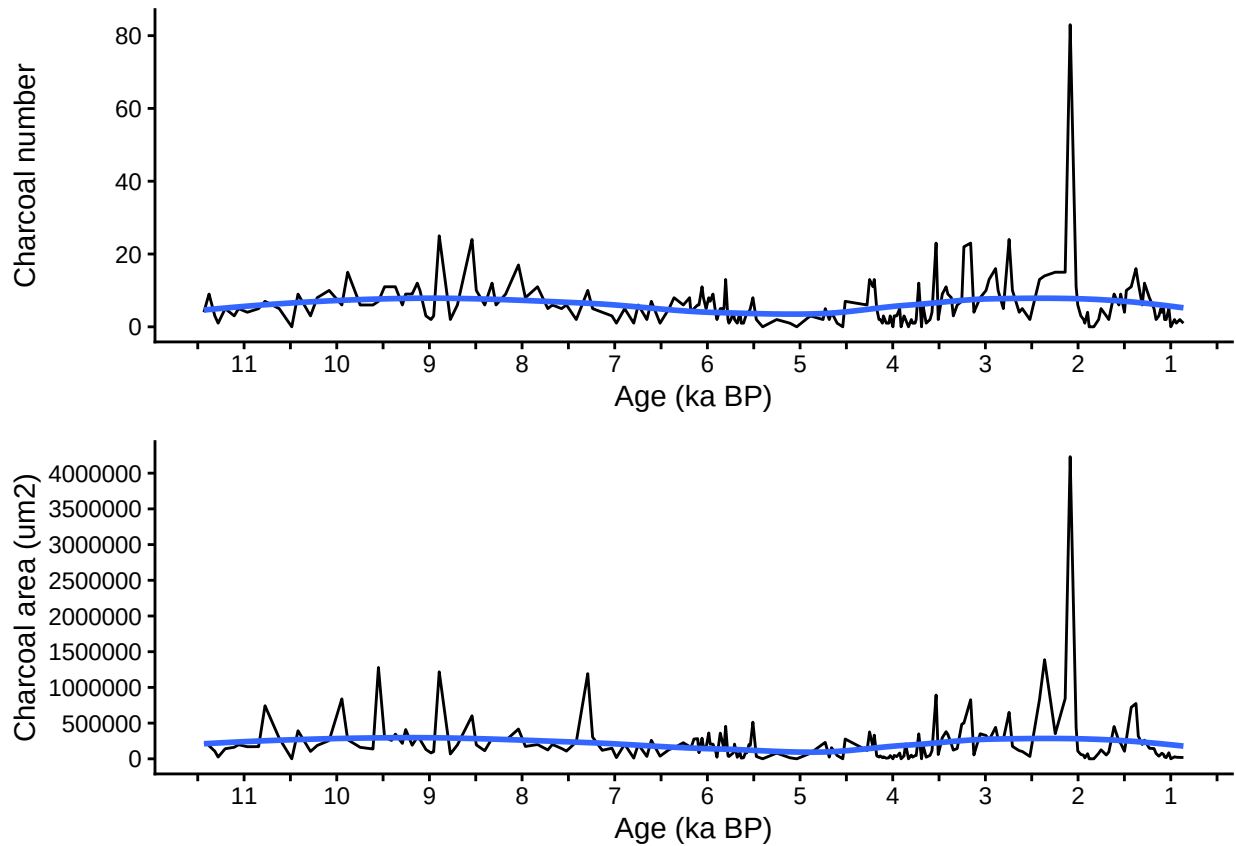


2.2.5. Charcoal number and area

```
CC.110 <- CharTotS.110.age %>%
  ggplot(aes(x = age/1000, y = CharCount)) +
  geom_line() +
  labs(x = "Age (ka BP)", y = "Charcoal number") +
  scale_x_reverse(breaks = seq(0.5, 12, by = 0.5),
    labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
  scale_y_continuous(breaks = seq(min(CharTotS.110.age$CharCount), max(CharTotS.110.age$CharCount), by =
  theme_classic() +
  geom_smooth(method = "loess", se = FALSE)

SCA.110 <- CharTotS.110.age %>%
  ggplot(aes(x = age/1000, y = sumCharArea)) +
  geom_line() +
  labs(x = "Age (ka BP)", y = "Charcoal area (um2)") +
  scale_x_reverse(breaks = seq(0.5, 12, by = 0.5),
    labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
  scale_y_continuous(breaks = seq(min(CharTotS.110.age$sumCharArea), max(CharTotS.110.age$sumCharArea),
  theme_classic() +
  geom_smooth(method = "loess", se = FALSE)
```

```
plot_grid(CC.110,SCA.110,ncol = 1, nrow = 2, align="v")
```



2.3. Data preparation for tapas and CharAnalysis

We need one file with the min, max, mean and median age, with the samples and also continuous depth (from 0 to 201)

```
Tr <- Tramacastilla_age %>%
  rename("median" = "age") %>%
  mutate(SampleId = str_replace_all(SampleId, "TRAM-", "TRAM20-1B-")) %>%
  left_join(tram20_depth_ages, by = c("SampleId", "median")) %>%
  filter(SampleId != 'NA') %>%
  dplyr::select(SampleId, min, max) %>%
  mutate(cmTop = seq(0, 201, 1)) %>%
  mutate(cmBot = seq(1, 202, 1)) %>%
  relocate(cmBot, .after = SampleId) %>%
  relocate(cmTop, .after = SampleId) %>%
  rename("AgeTop" = min) %>%
  rename("AgeBot" = max) %>%
  mutate(across(everything(), ~ str_replace(., "^TRAM20-1B-", "TRAM-"))) %>%
  merge(CharTotS.110.age, by = "SampleId") %>%
  rename("char" = CharCount) %>%
  rename("chararea" = sumCharArea) %>%
  dplyr::select(cmTop, cmBot, AgeTop, AgeBot, char, chararea) %>%
```

```
mutate(across(everything(), as.numeric)) %>%
arrange(cmTop)
```

We need another file with cmTop, cmBot, AgeTop, AgeBot, CharVol and CharCount (can be either counting or charcoal area)

```
head (mass.density)#dataframe with volume and mass information of charcoal samples
```

```
##      SampleId Density..wet.sample...g.cm3. Density..dry.sample...g.cm3.
## 1 TRAM-3U-19-20                1.443714                0.7731429
## 2 TRAM-3U-20-21                1.352385                0.7143846
## 3 TRAM-3U-21-22                1.312714                0.6542143
## 4 TRAM-3U-22-23                1.388818                0.6557727
## 5 TRAM-3U-23-24                1.175850                0.4662500
## 6 TRAM-3U-24-25                1.226727                0.4925000
##      Mass..g.
## 1      1.5025
## 2      1.6349
## 3      1.6743
## 4      1.5234
## 5      1.5846
## 6      1.5901
```

```
volume <- mass.density %>%
  mutate(vol = Mass..g./Density..wet.sample...g.cm3.) %>%
  dplyr::select(SampleId, vol)

CharAnalysis.tram.110 <- Tramacastilla_age %>%
  rename("median" = "age") %>%
  mutate(SampleId = str_replace_all(SampleId, "TRAM-", "TRAM20-1B-")) %>%
  left_join(tram20_depth_ages, by = c("SampleId", "median")) %>%
  relocate(SampleId, .before = composite_depth) %>%
  filter(SampleId != 'NA') %>%
  dplyr::select(SampleId, min, max) %>%
  mutate(cmTop = seq(0, 201, 1)) %>%
  mutate(cmBot = seq(1, 202, 1)) %>%
  relocate(cmBot, .after = SampleId) %>%
  relocate(cmTop, .after = SampleId) %>%
  rename("AgeTop" = min) %>%
  rename("AgeBot" = max) %>%
  mutate(across(everything(), ~ str_replace(., "^TRAM20-1B-", "TRAM-"))) %>%
  merge(CharTotS.110.age, by = "SampleId") %>%
  rename("char" = CharCount) %>%
  rename("chararea" = sumCharArea) %>%
  full_join(volume, by = "SampleId") %>%
  dplyr::select(cmTop, cmBot, AgeTop, AgeBot, char, chararea, vol) %>%
  mutate(across(everything(), as.numeric)) %>%
  arrange(cmTop)
```

2.4. Perform tapas

2.4.1. Explore data

Check data and treat it. Any gaps? continuous sampling? duplicates?

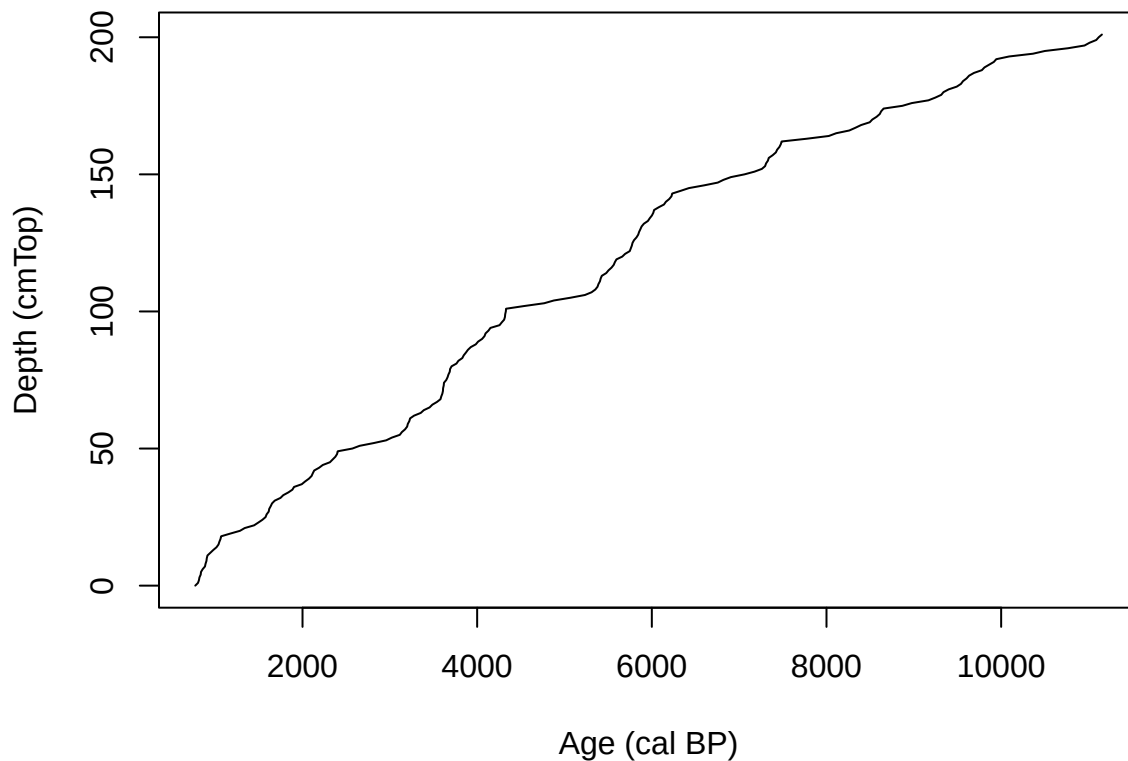
```
Tr <- tapas::check_pretreat(Tr)
```

```
## [1] "No slump samples detected."
## [1] ""
## [1] "No gaps found in the depth scale"
## [1] ""
## [1] "No overlapping depths found"
## [1] ""
## [1] "Warning: the age scale is not continuous."
```

```
Tr_i <- tapas::pretreatment_data(series = Tr)
```

Plot the depth-age relationship

```
par(mfrow = c(1,1), mar = c(5,5,2,2))
plot(Tr$AgeTop, Tr$cmTop, type = "l",
     xlab = "Age (cal BP)",
     ylab = "Depth (cmTop)")
```

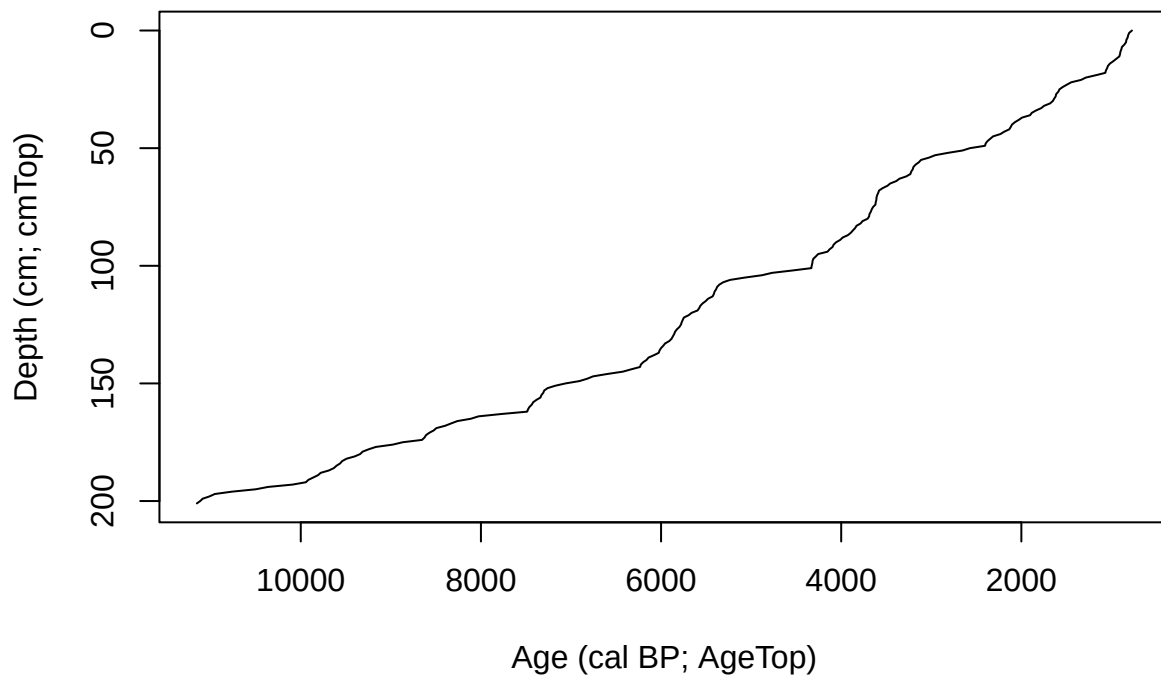


Set x-axis and y-axis limits, and the direction of the axes and replot the depth-age relationship

```

x_lim <- c(max(Tr$AgeTop), min(Tr$AgeTop))
y_lim <- c(max(Tr$cmTop), min(Tr$cmTop))
plot(Tr$AgeTop, Tr$cmTop, type = "l",
     xlim = x_lim, ylim = y_lim,
     xlab = "Age (cal BP; AgeTop)",
     ylab = "Depth (cm; cmTop)")

```

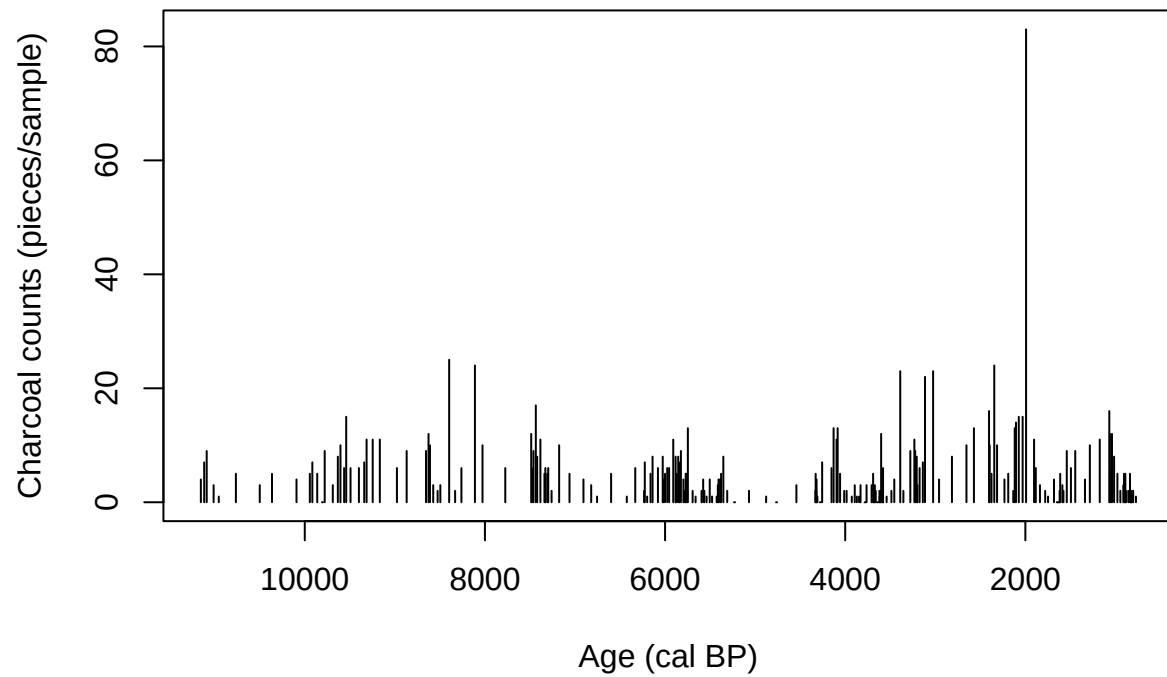


Plot the raw data: charcoal counts against age

```

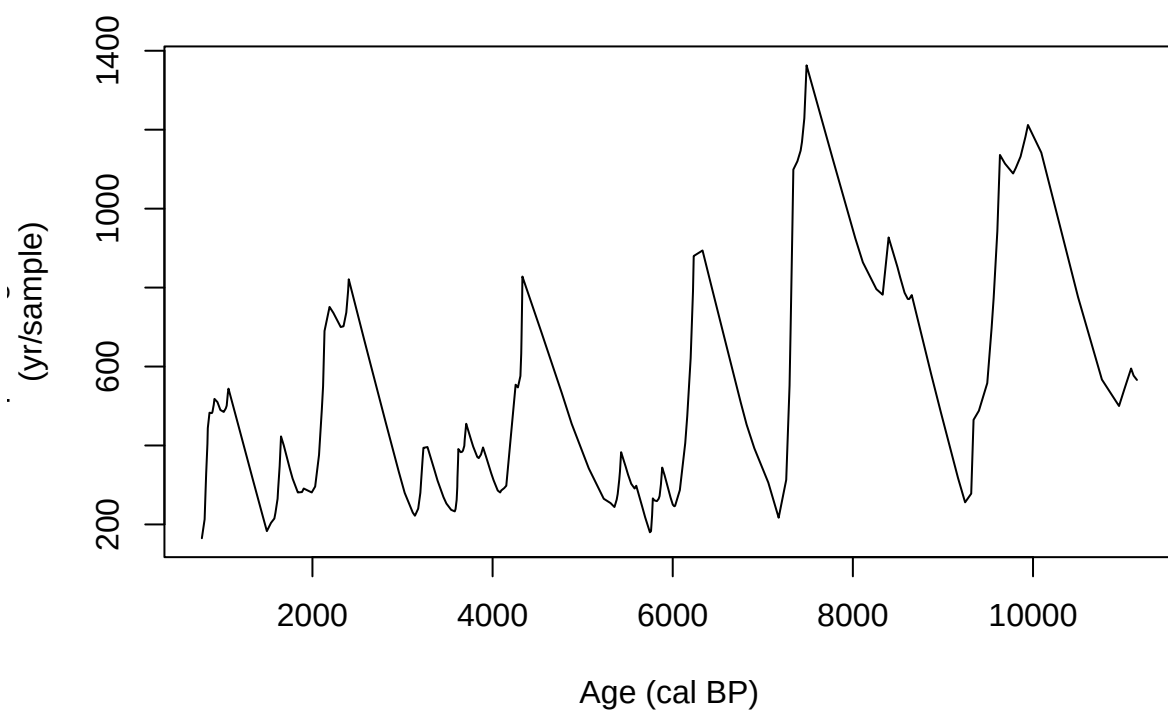
plot(Tr$AgeTop, Tr$char, type = "h",
     xlim = x_lim,
     xlab = "Age (cal BP)",
     ylab = "Charcoal counts (pieces/sample)")

```



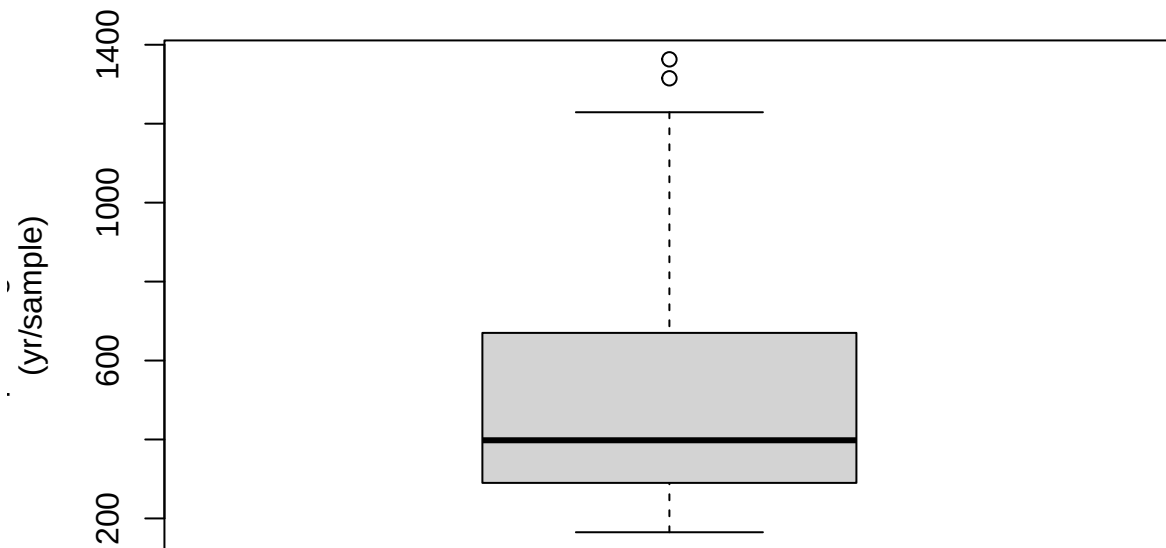
Calculate and plot the sampling resolution (sample integration time) for the record (yr/sample)

```
yr_smpl <- Tr$AgeBot - Tr$AgeTop
plot(Tr$AgeTop, yr_smpl, type = "l",
     xlab = "Age (cal BP)",
     ylab = "Sample integration time\n(yr/sample)")
```



Check the distribution of sampling resolution values

```
boxplot(yr_smpl, ylab = "Sample integration time\n(yr/sample)")
```

```
summary(yr_smpl)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  165.0   290.2   398.0   502.2   665.0  1363.0
```

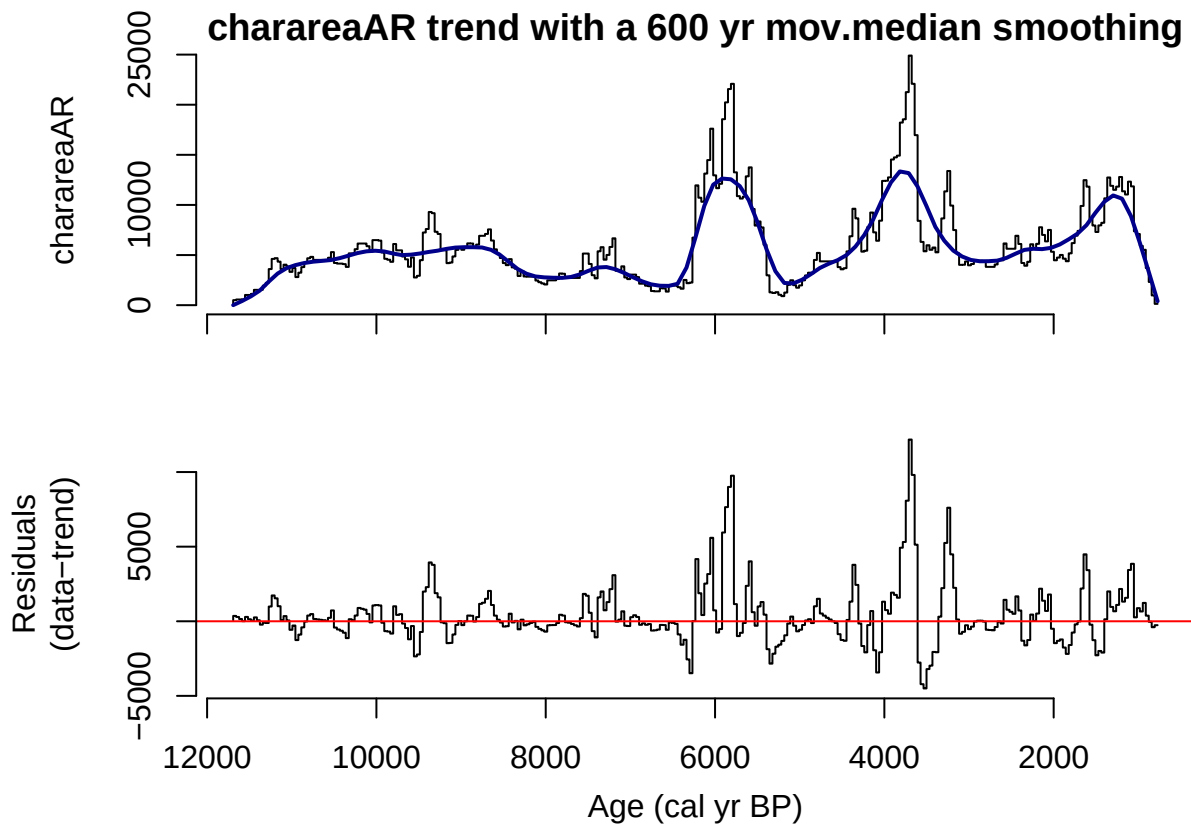
2.4.2. CharAnalysis-type with 'local threshold'

Bin the data

```
tr_i <- tapas::pretreatment_data(series = Tr, out = "accI",
                                first = 772, last = 11720,
                                yrInterp = 35)
```

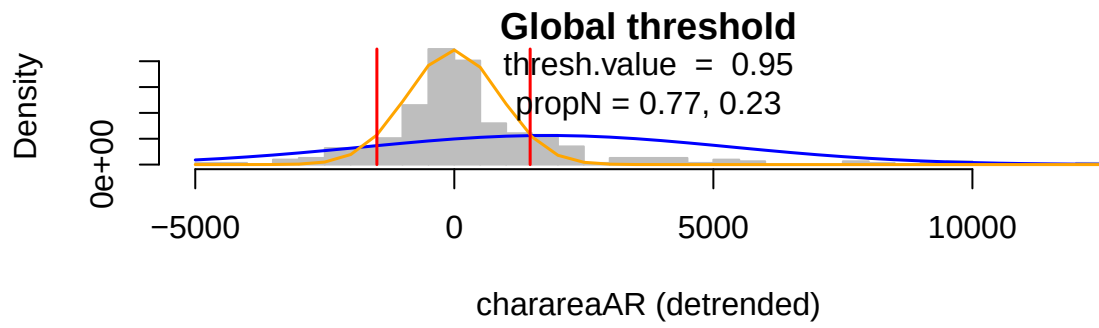
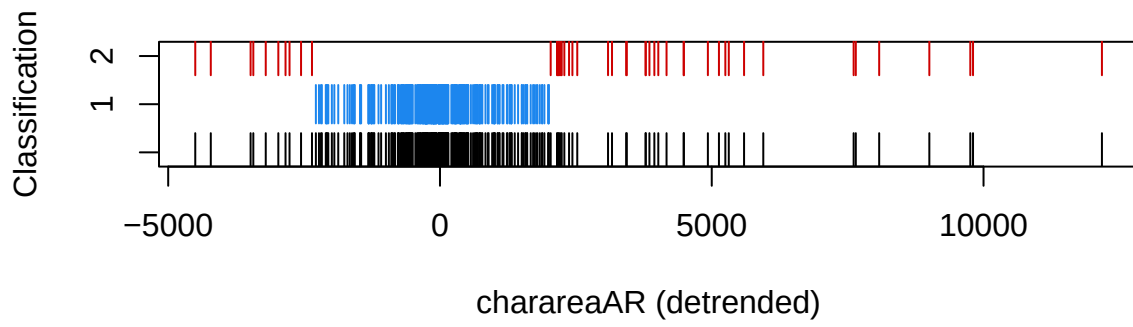
Detrend the data

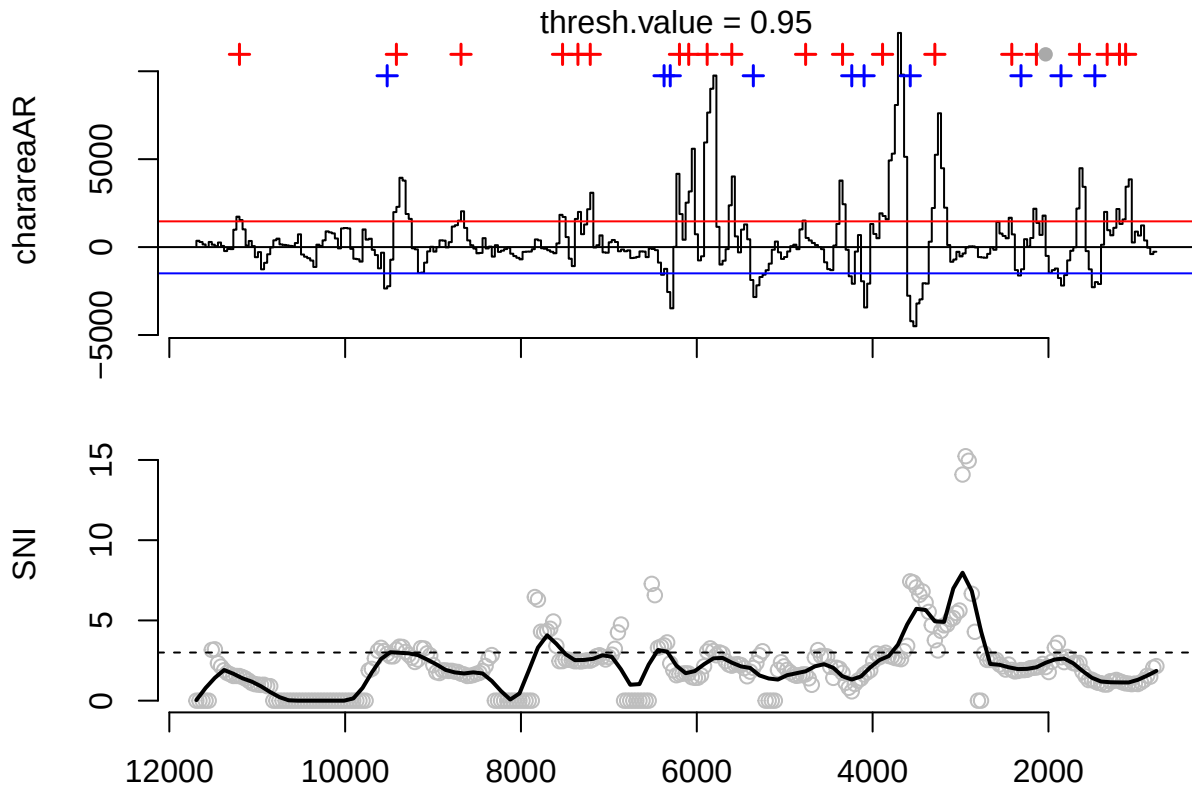
```
tr_detr <- tapas::SeriesDetrend(tr_i, detr.type = "mov.median",
                                smoothing.yr = 600)
```



Decompose noise & potential peaks

```
tr_glob <- tapas::global_thresh(tr_detr, proxy = "charareaAR",  
                                thresh.value = 0.95)
```



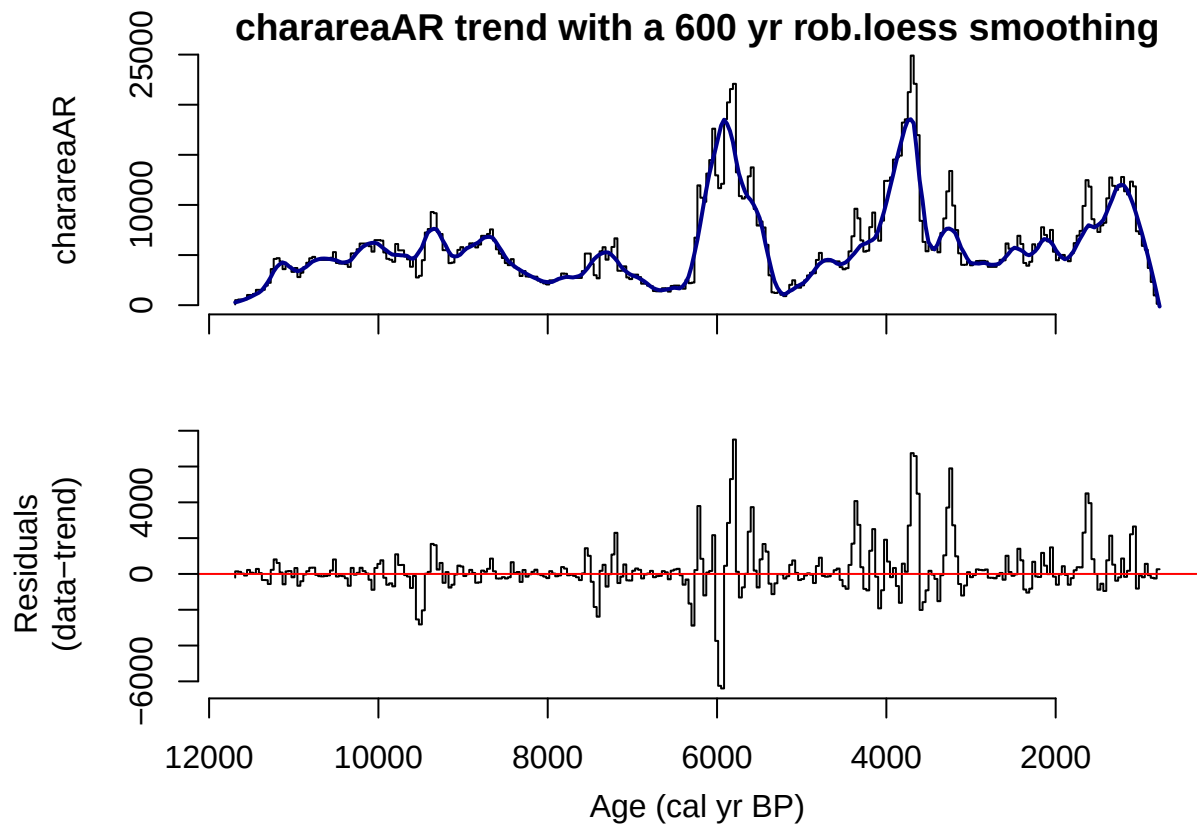


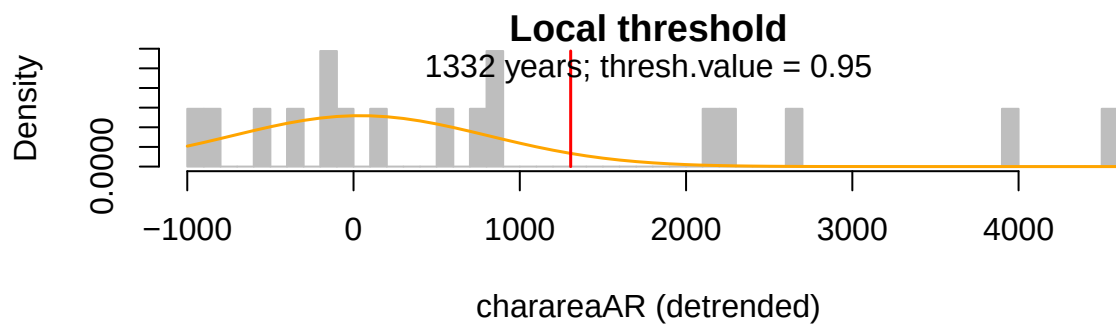
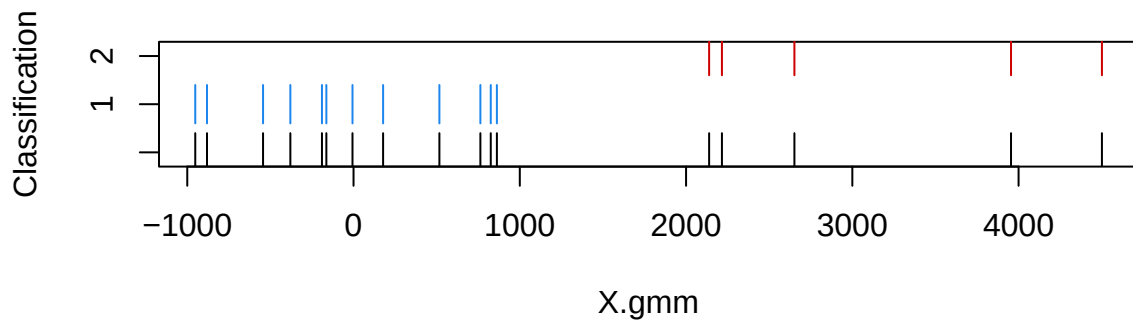
2.4.3. Evaluate the results

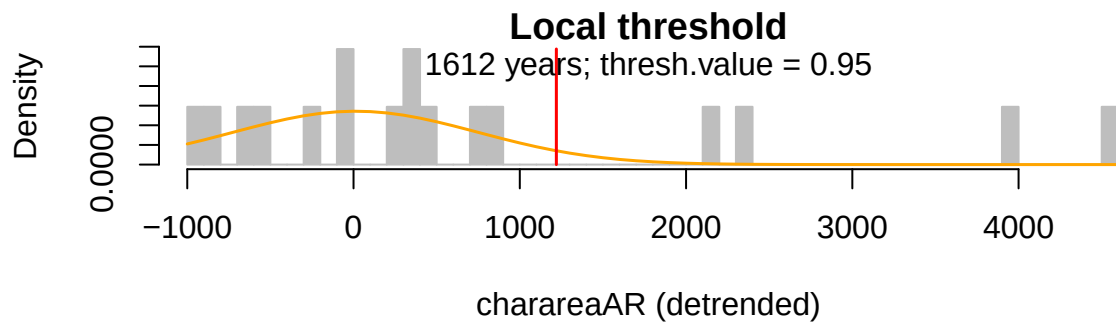
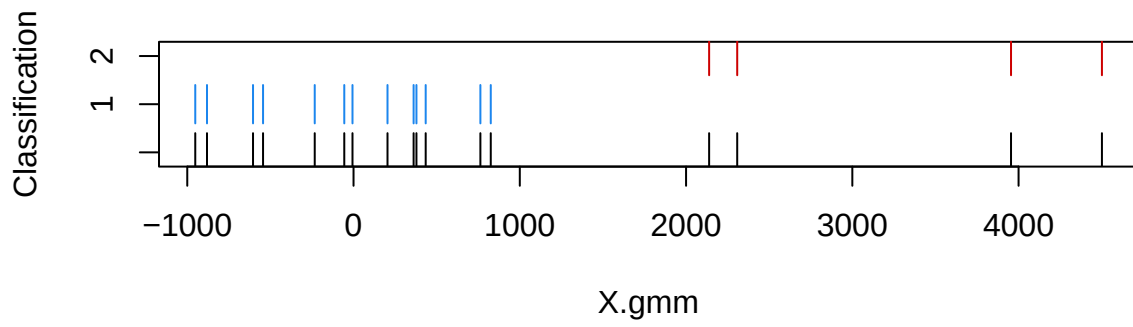
Check SNI plot produced with previous function (the 2nd to last plots). Evaluate sensitivity to smoothing-window widths using the `peak_detection()` wrapper function

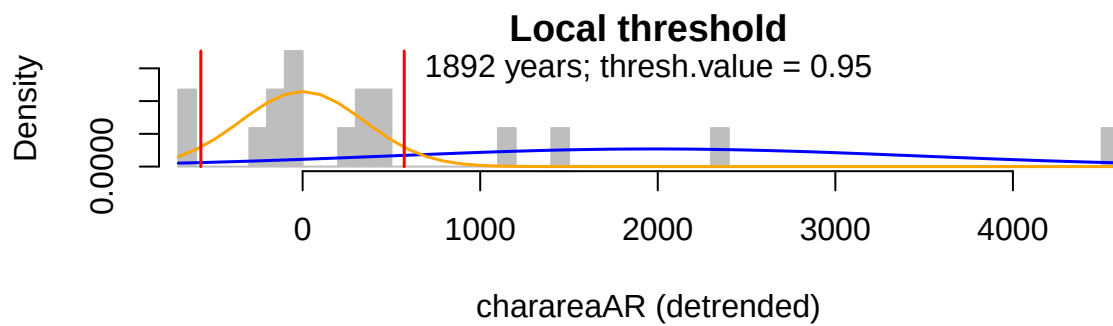
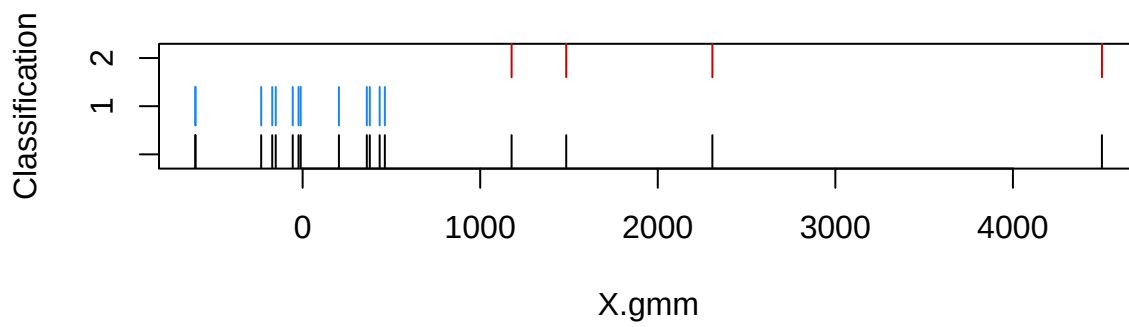
```
tr_loc_prev <- tapas::peak_detection(series = Tr, proxy = "chararea",
  first = 772, last = 11720,
  yrInterp = 35,
  thresh_type = "local",
  detr_type = "rob.loess",
  smoothing_yr = 600,
  thresh_value = 0.95,
  sens = T,
  smoothing_yr_seq = c(500, 600, 700,
    800, 900, 1000))
```

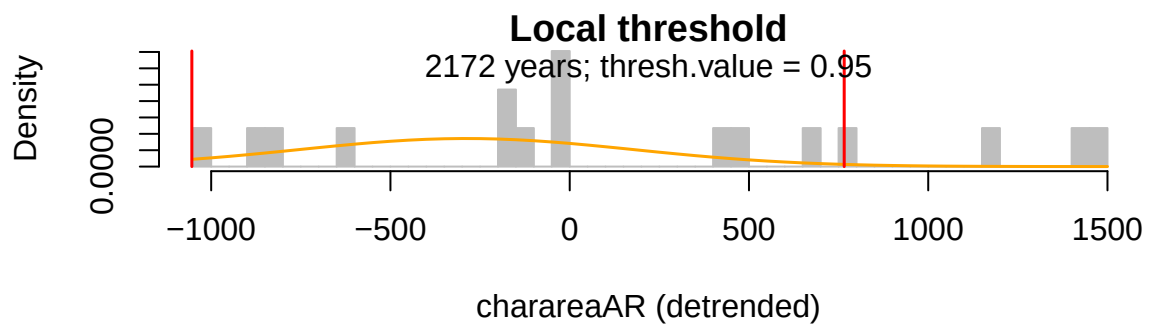
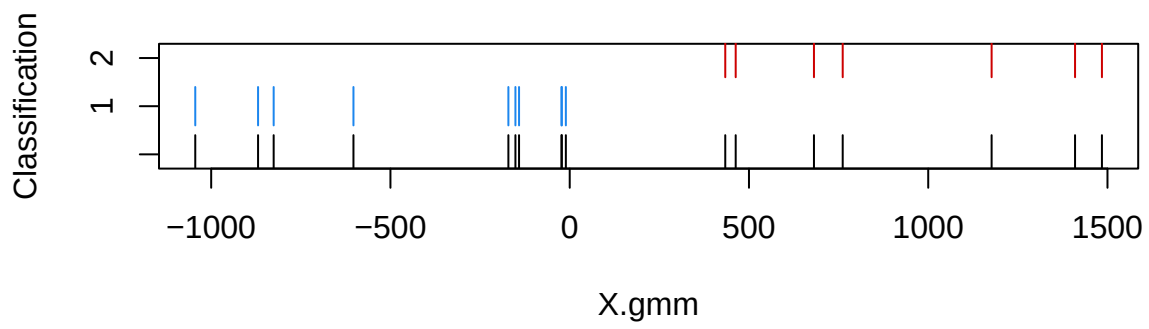
```
## [1] "No slump samples detected."
## [1] ""
## [1] "No gaps found in the depth scale"
## [1] ""
## [1] "No overlapping depths found"
## [1] ""
## [1] "Warning: the age scale is not continuous."
```

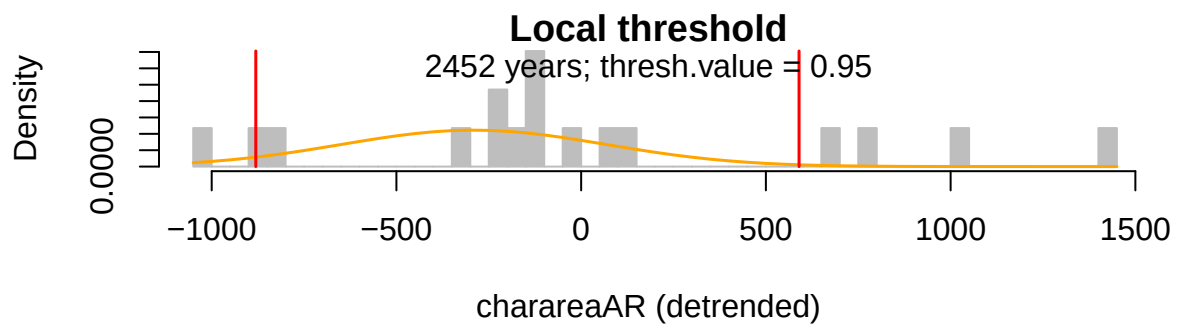
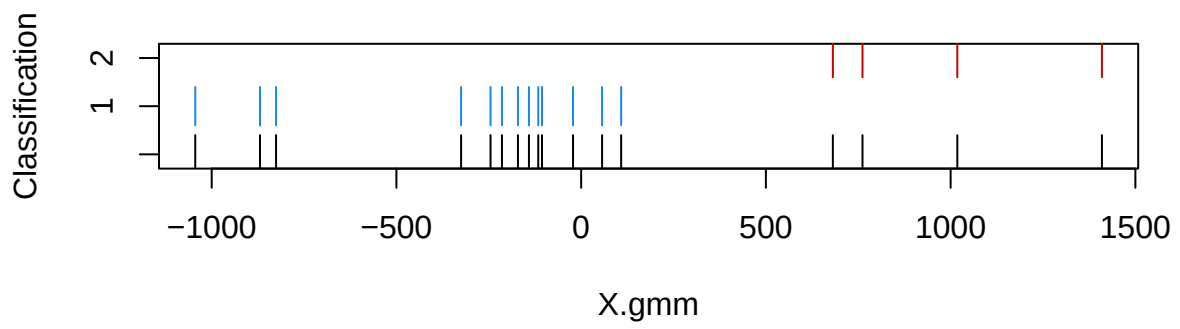


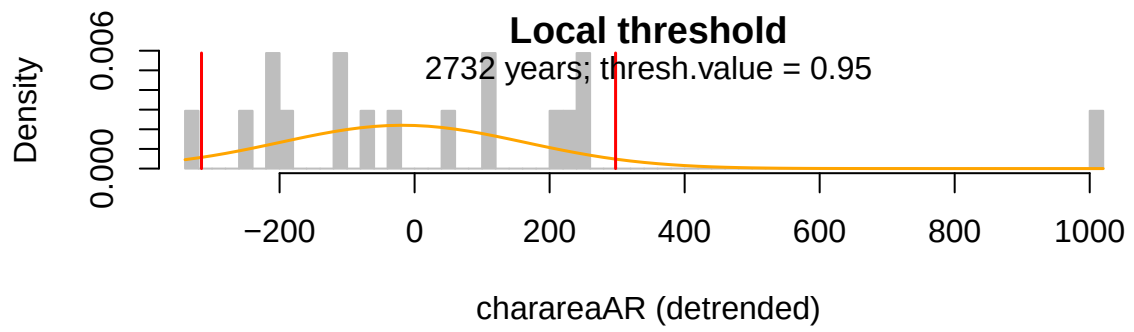
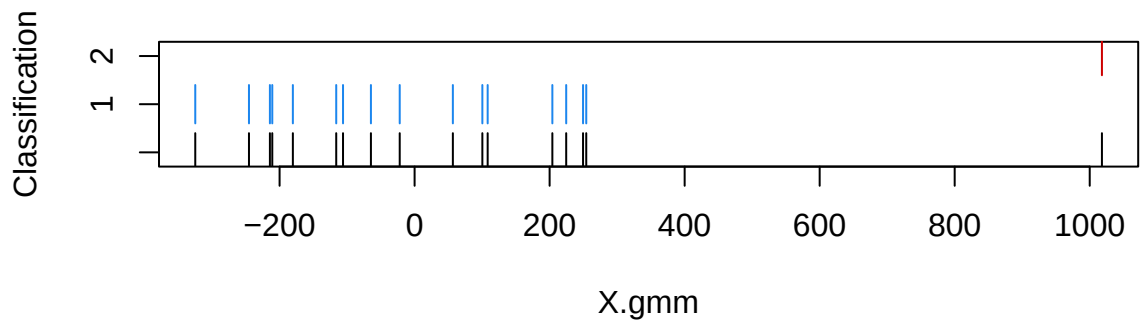


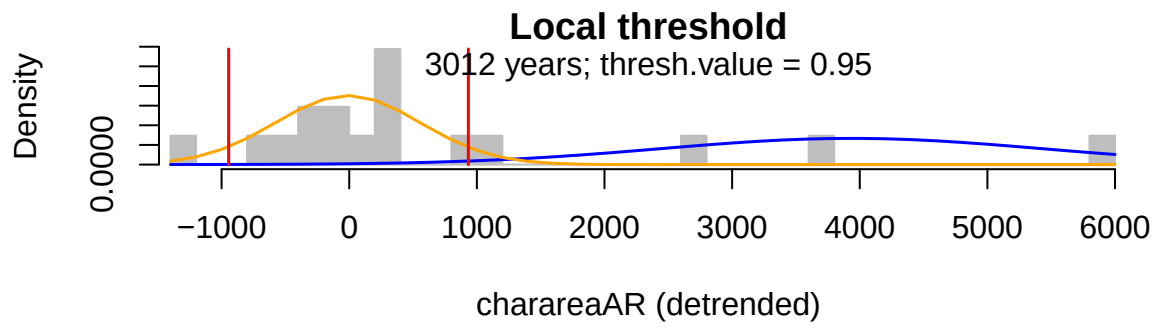
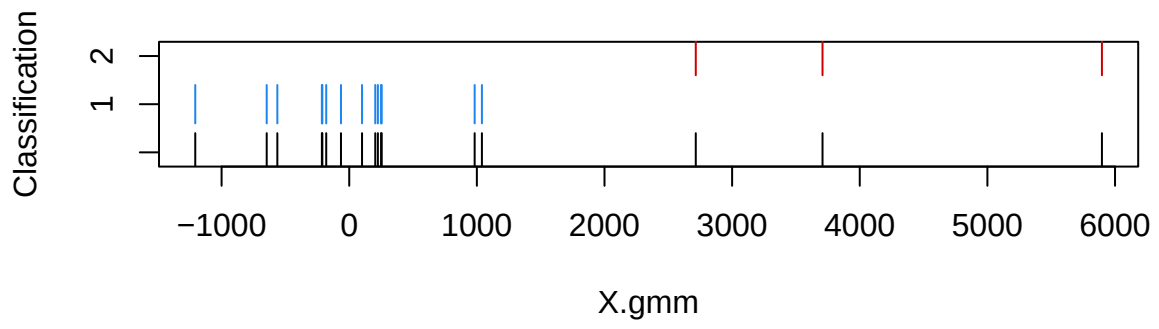


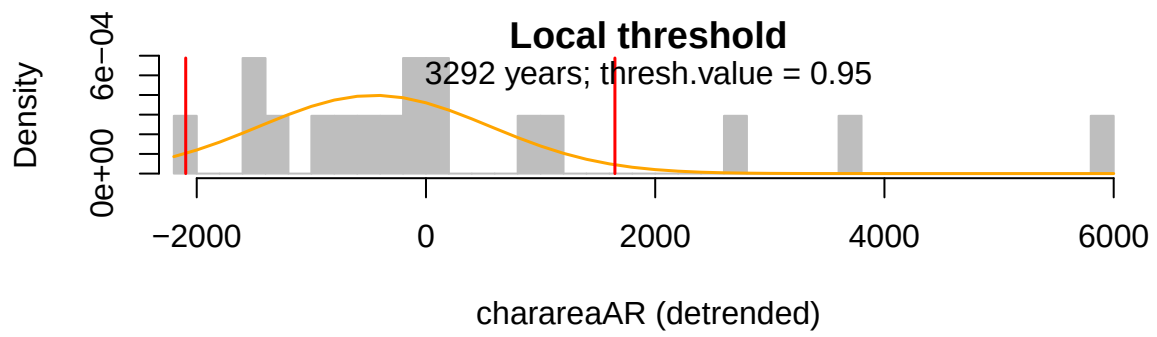
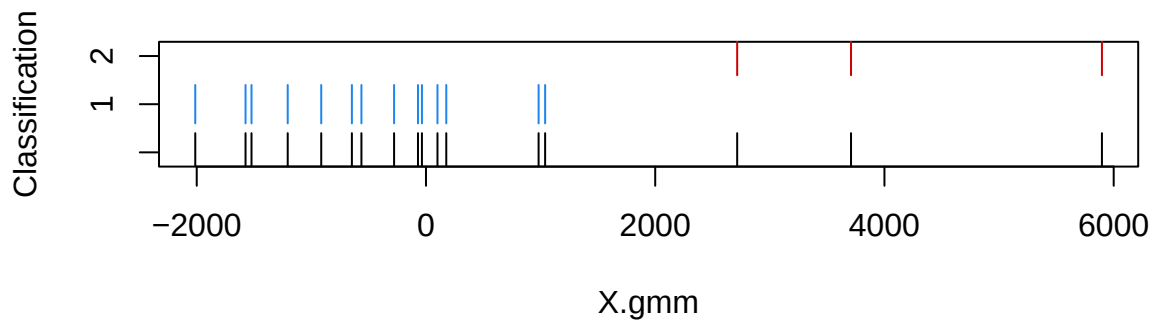


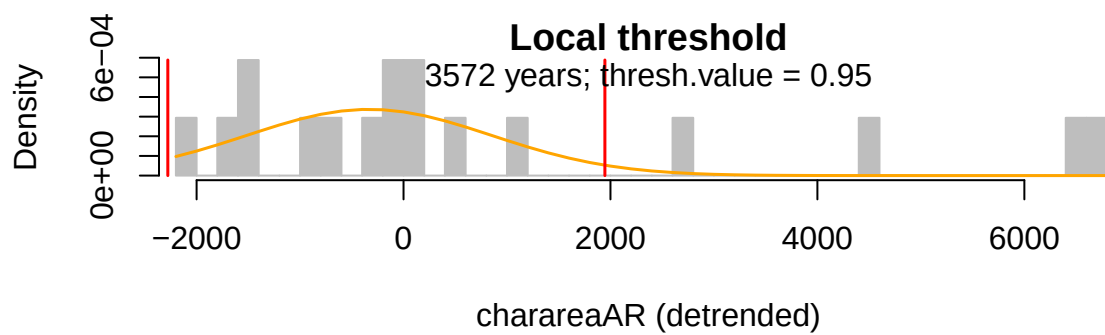
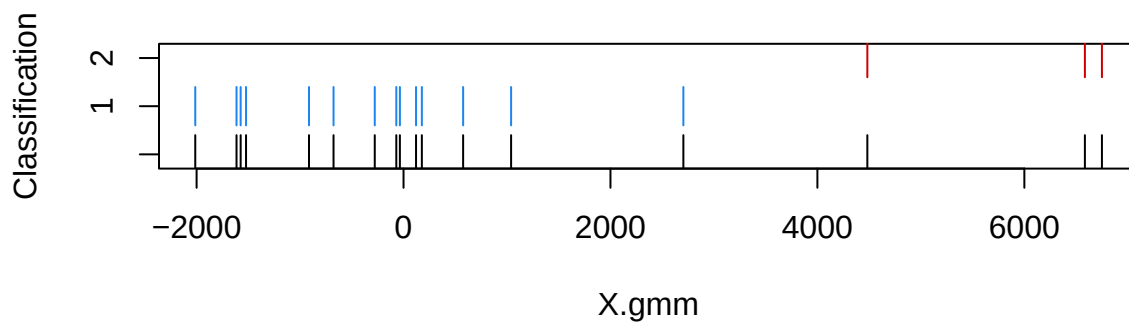


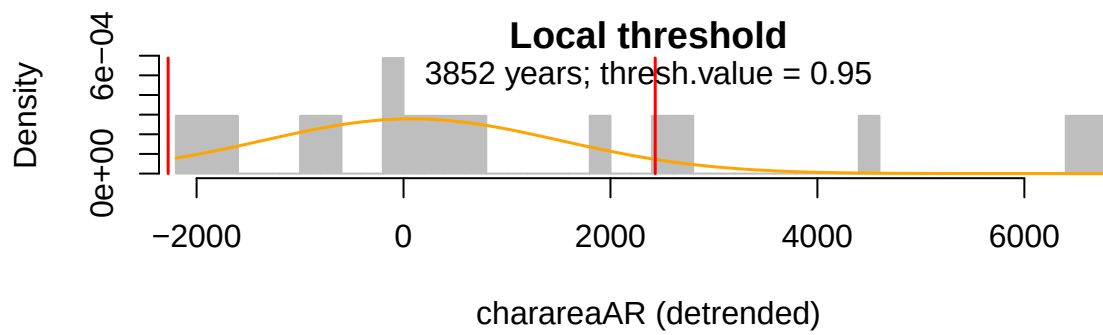
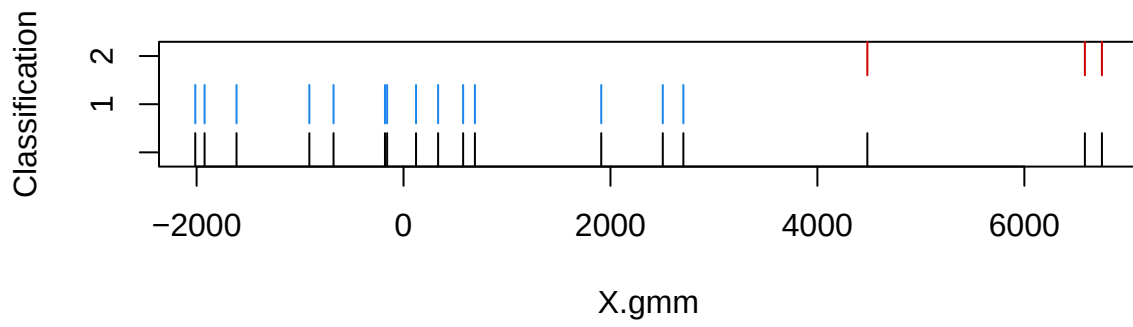


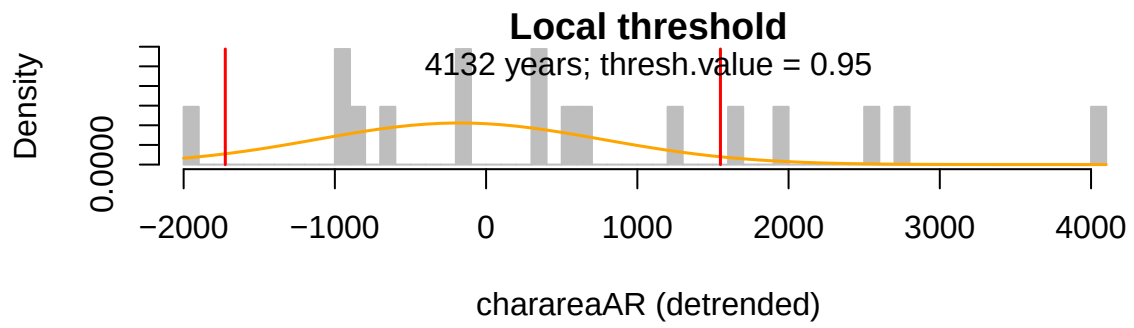
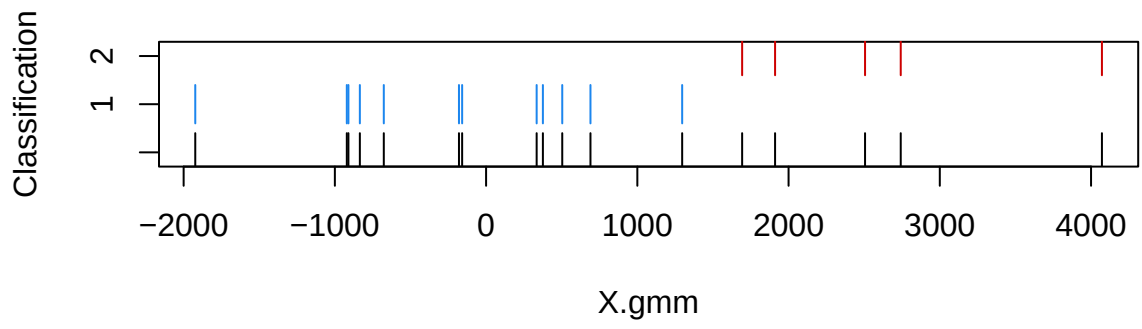


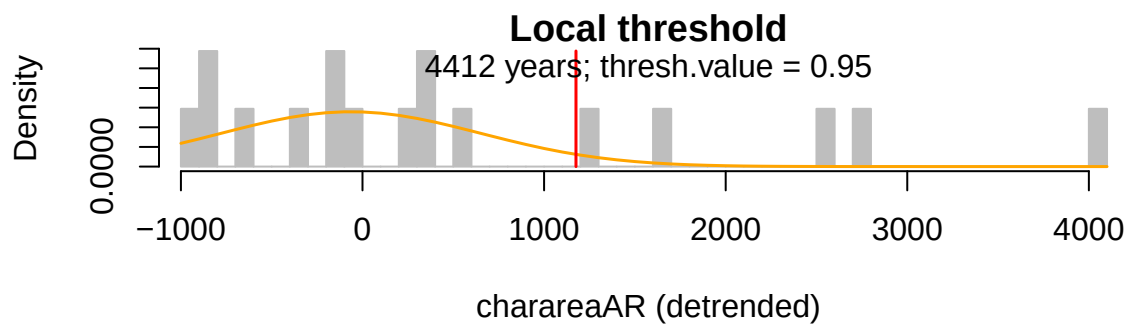
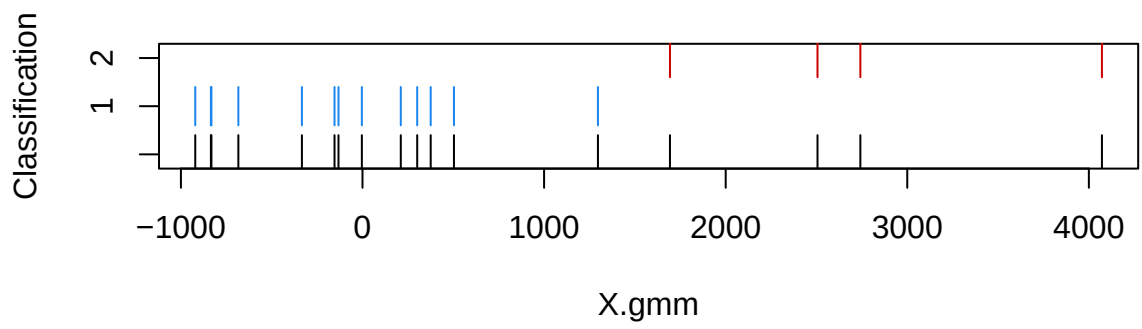


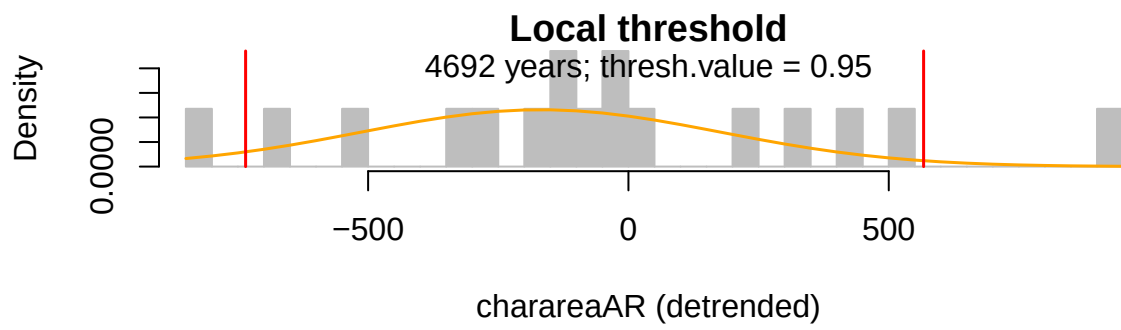
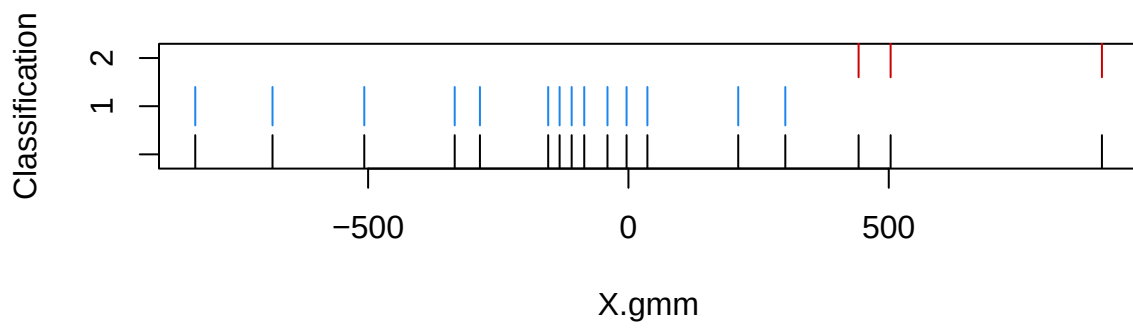


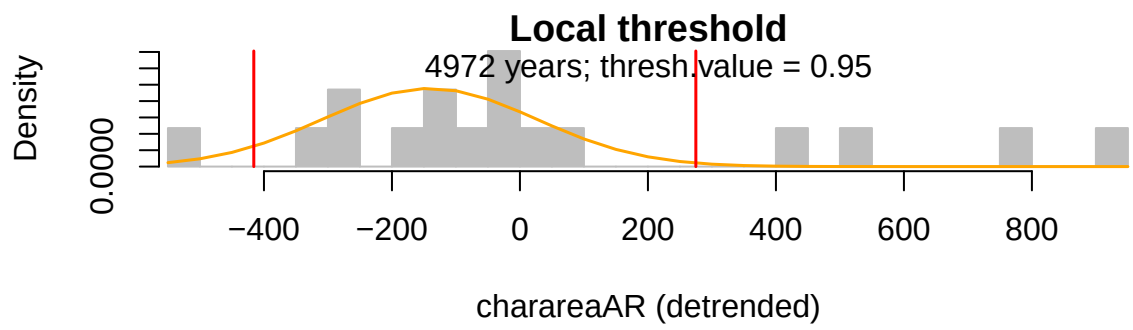
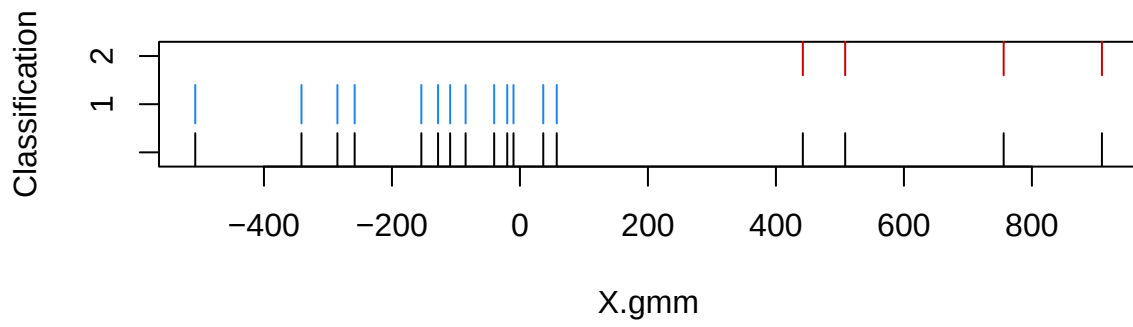


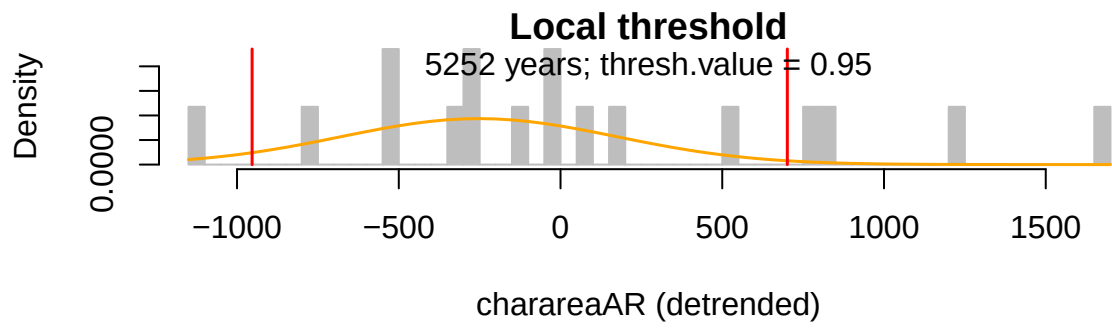
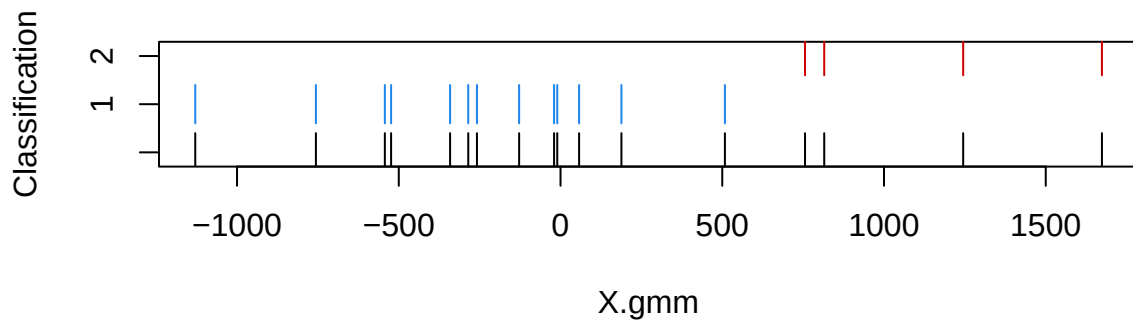


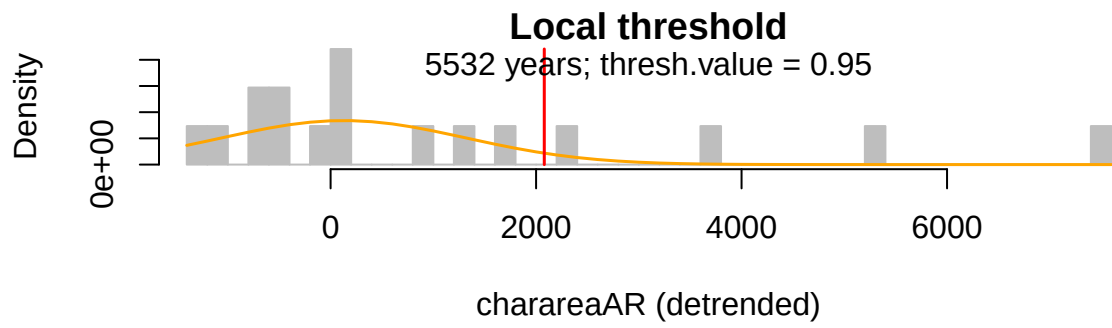
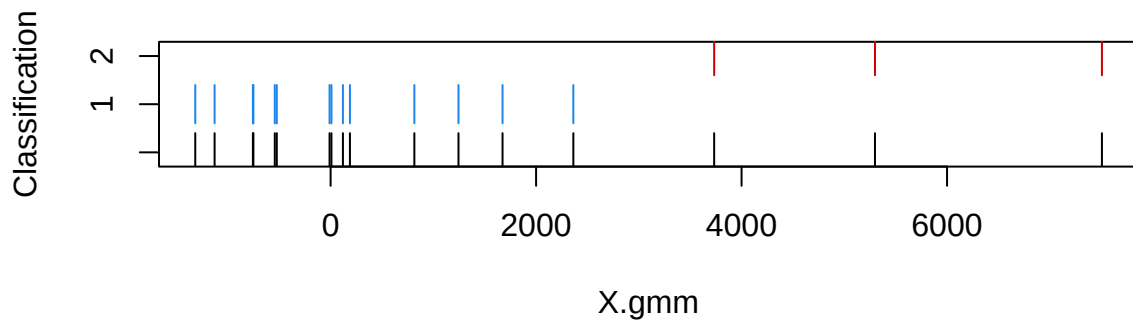


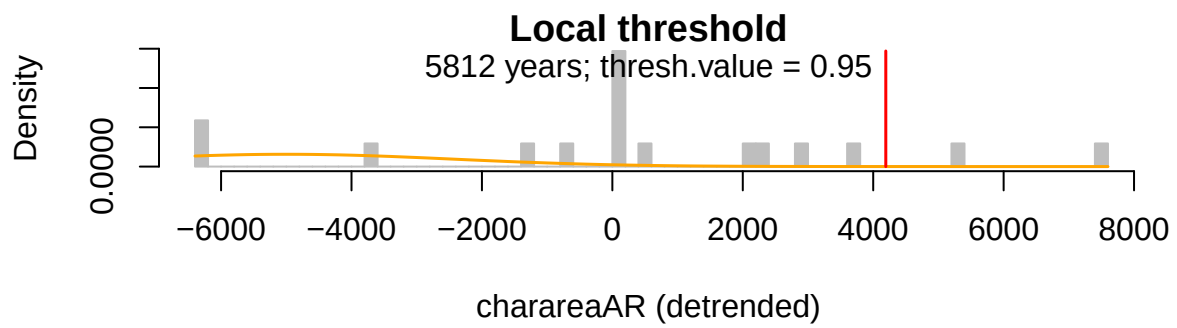
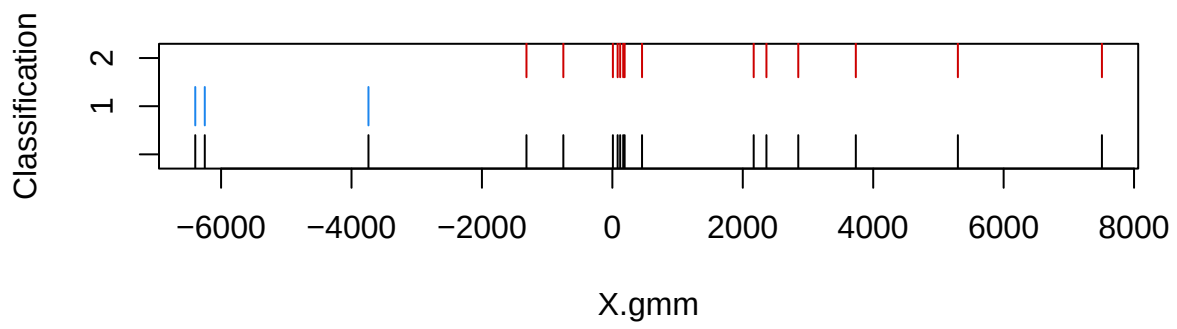


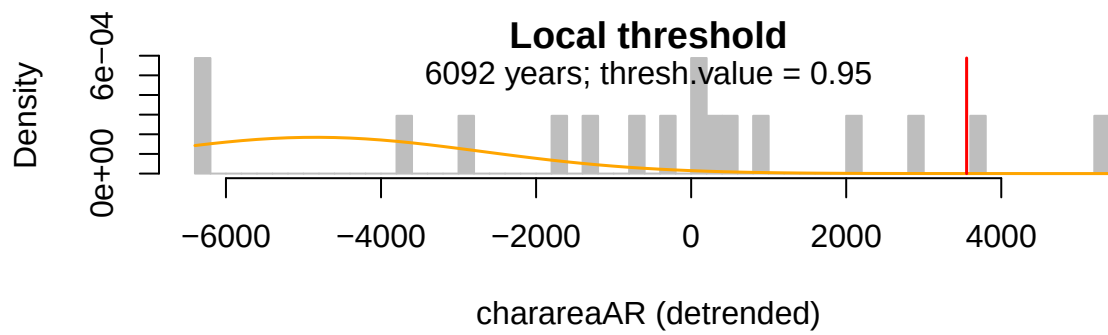
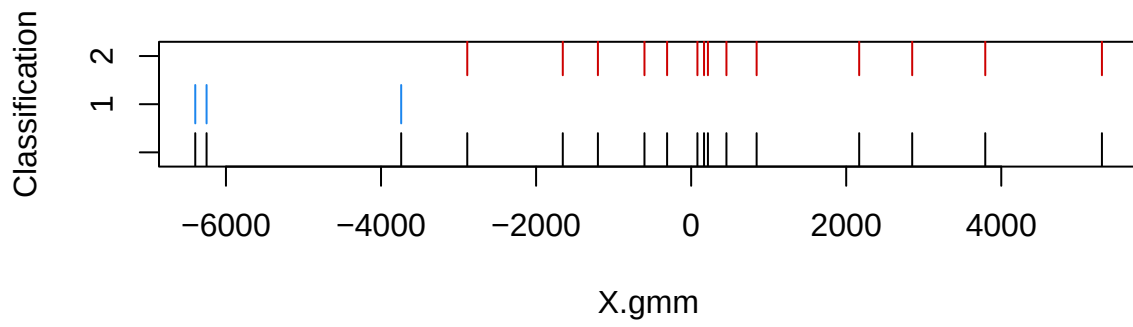


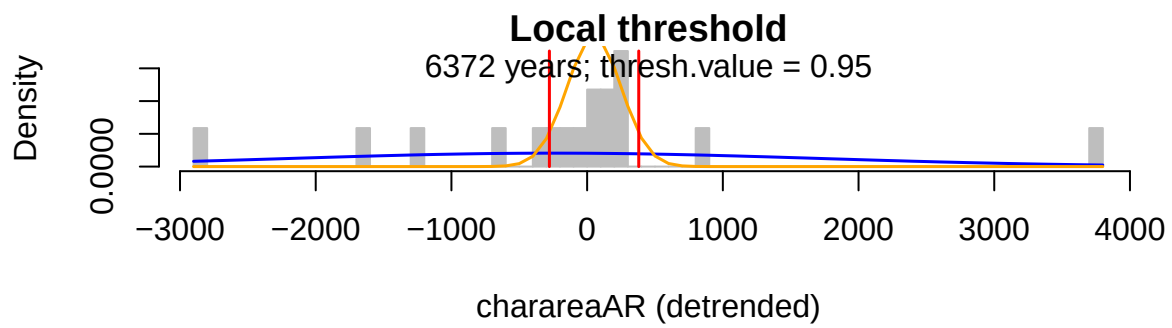
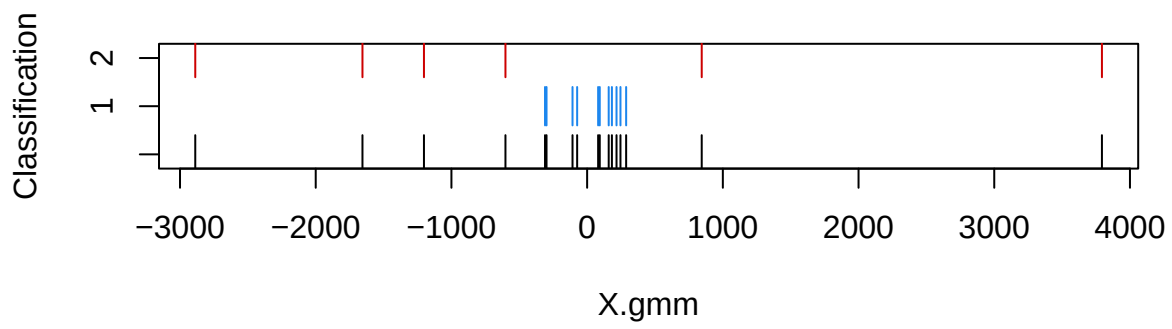


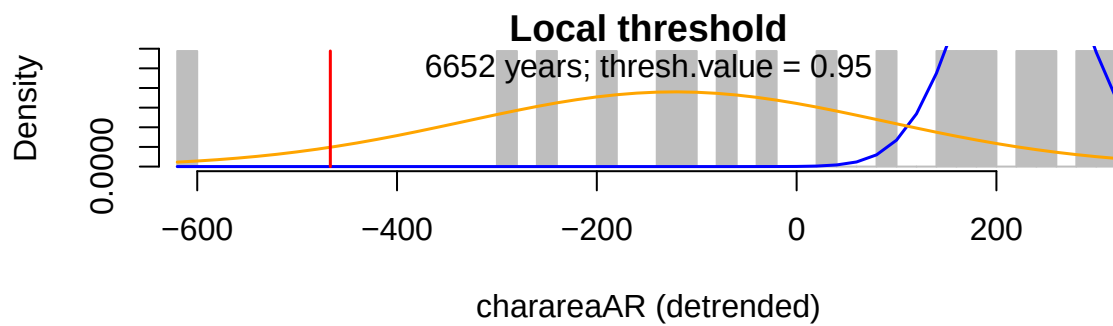
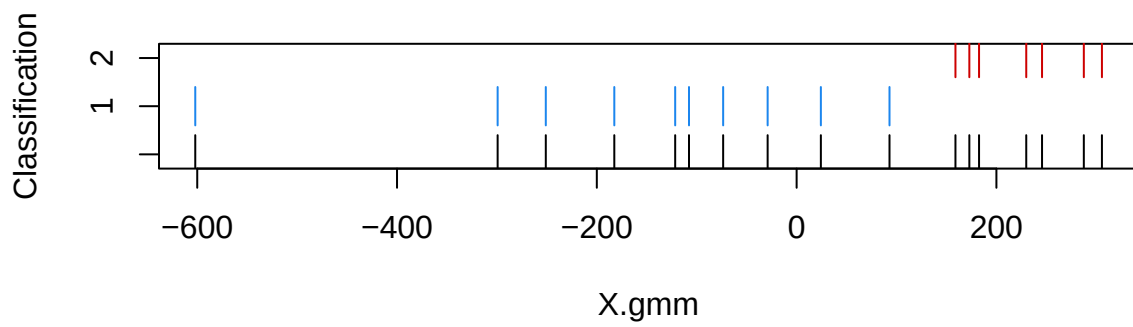


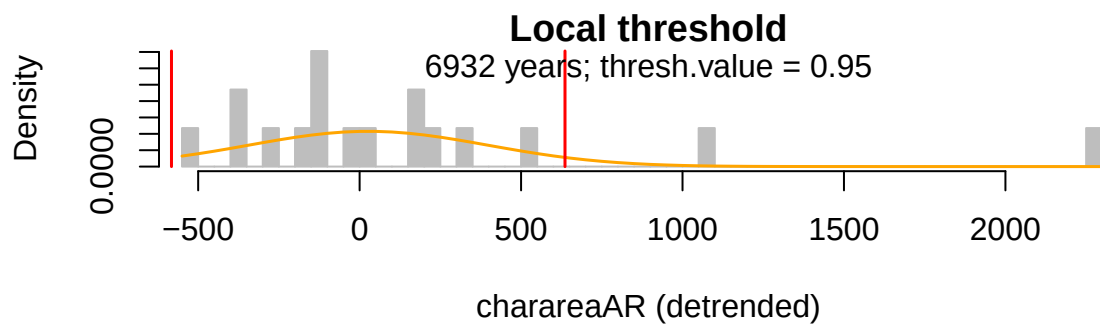
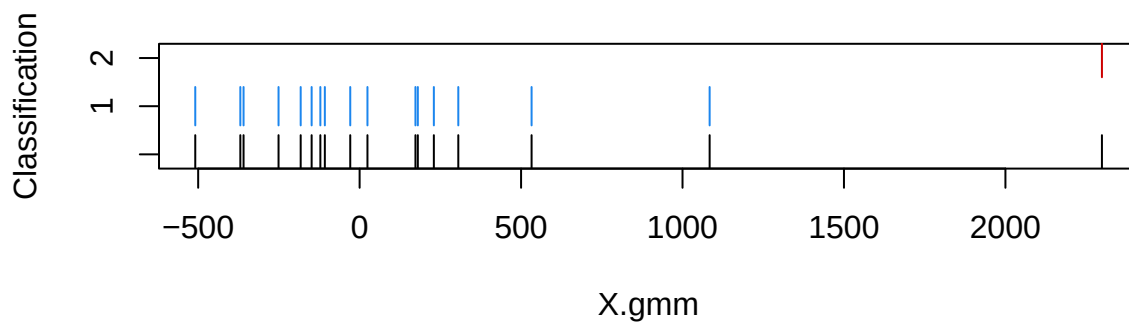


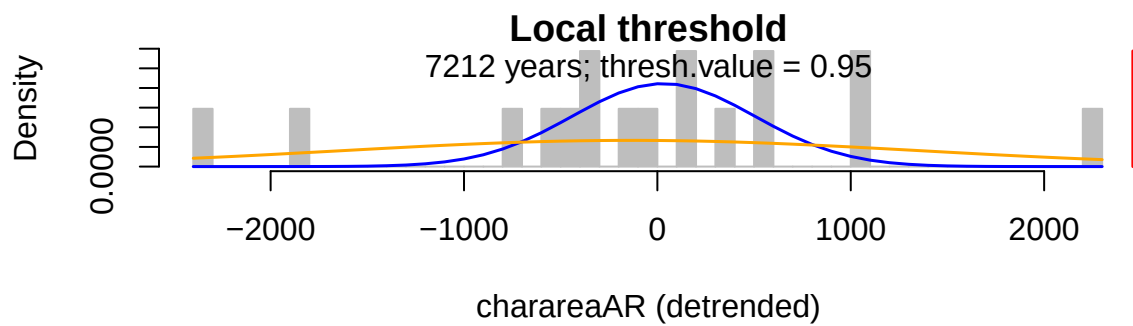
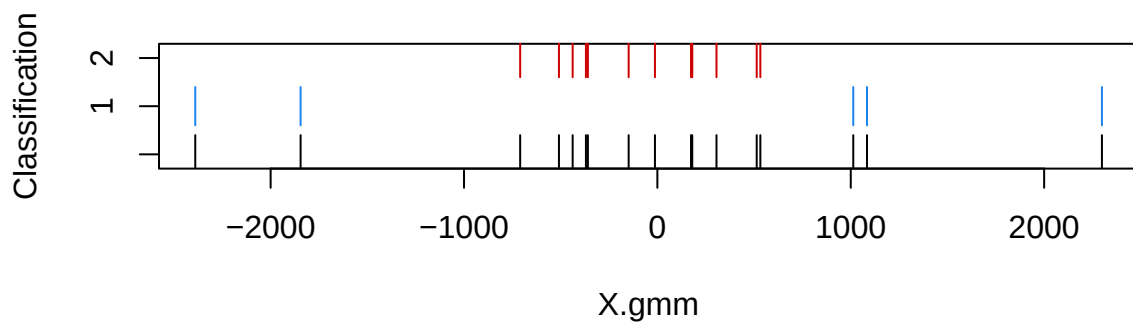


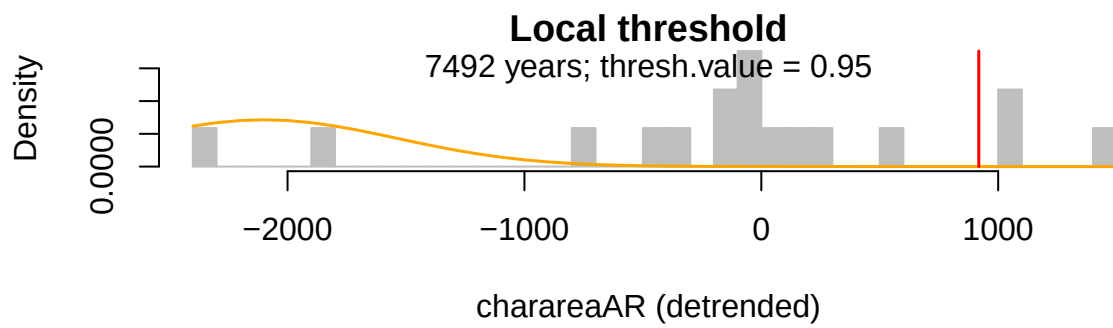
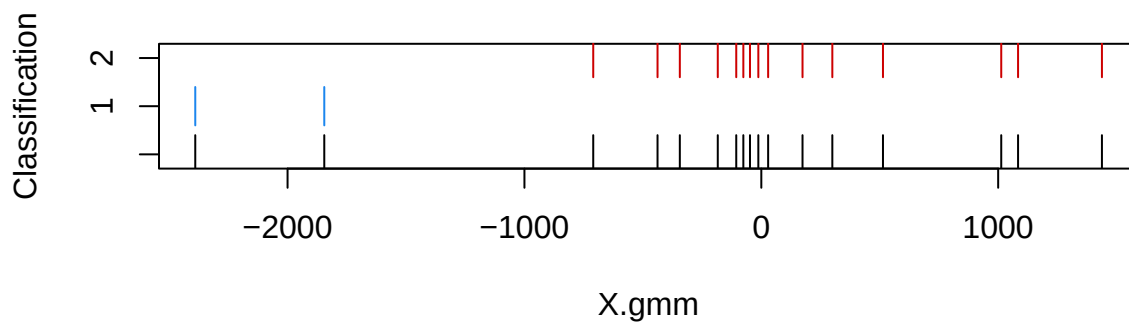


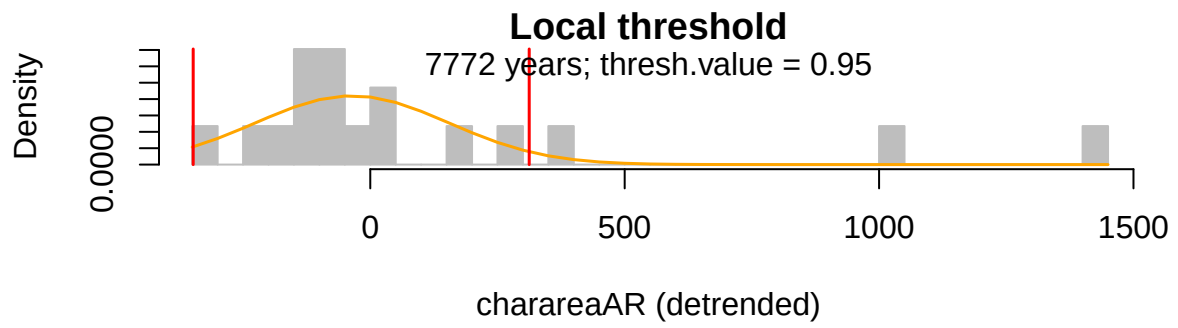
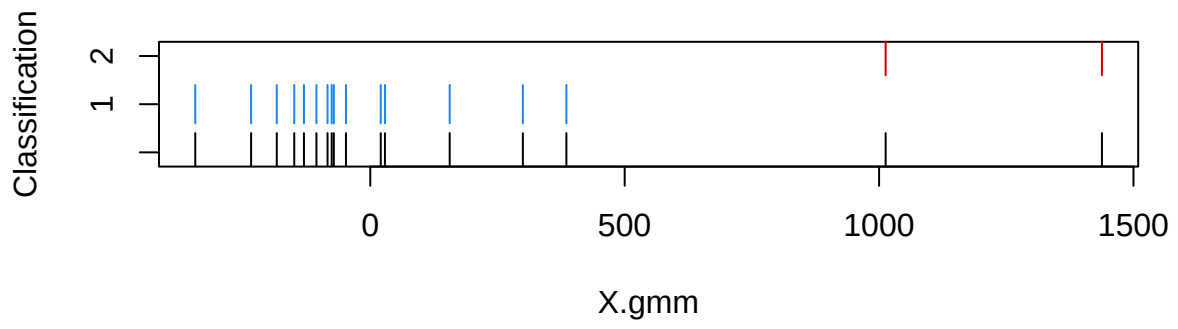


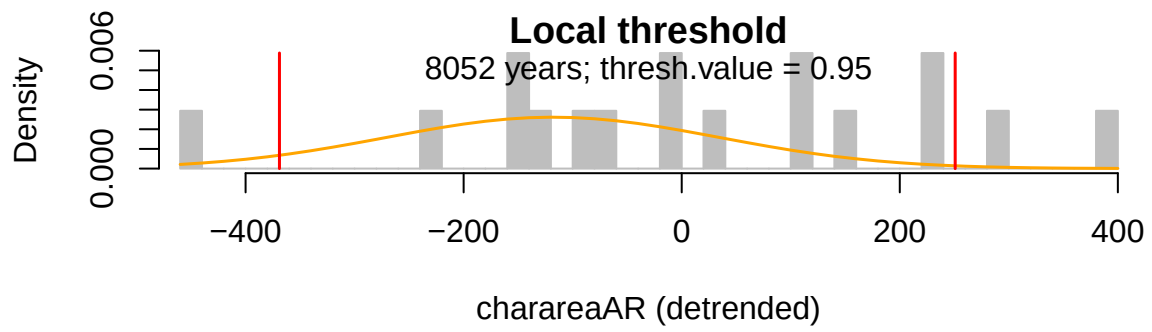
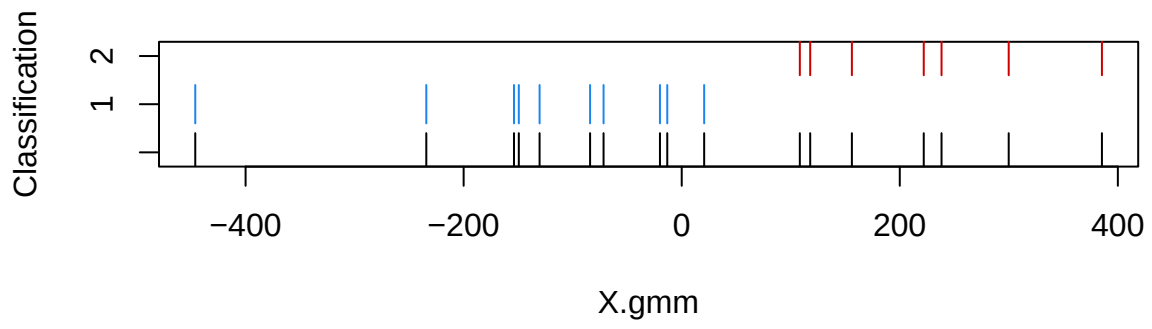


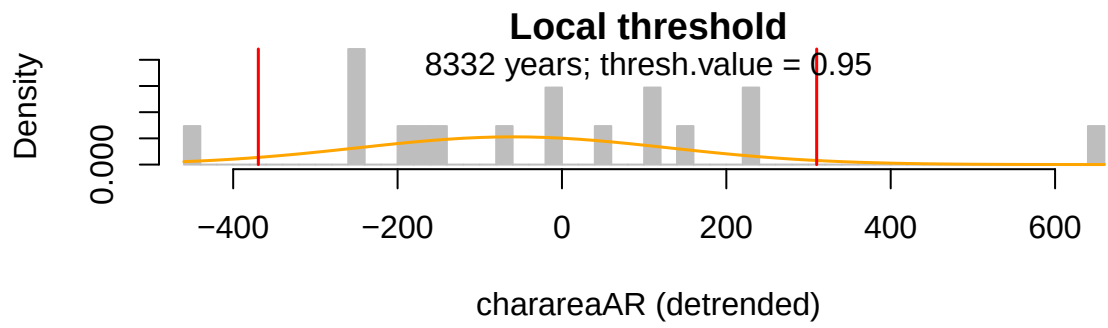
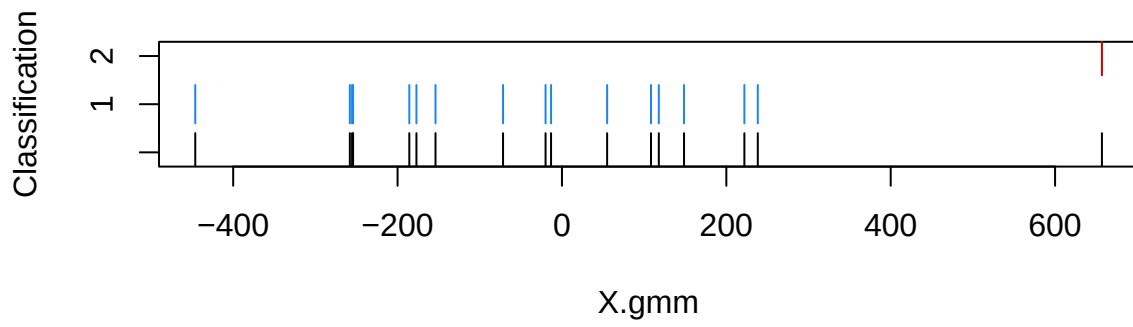


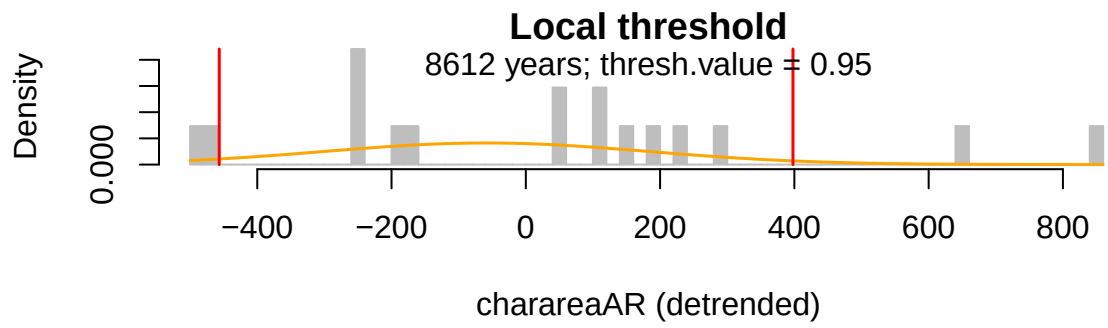
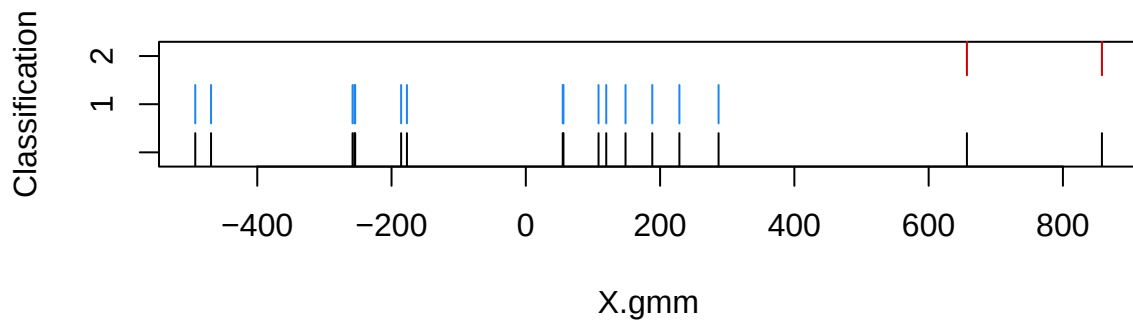


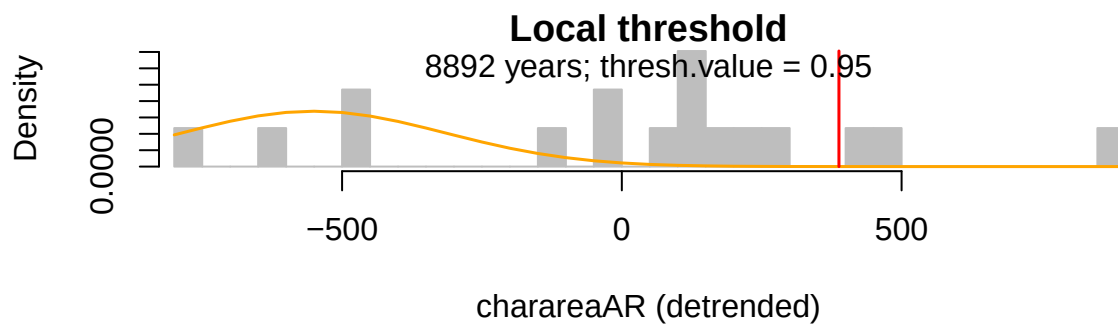
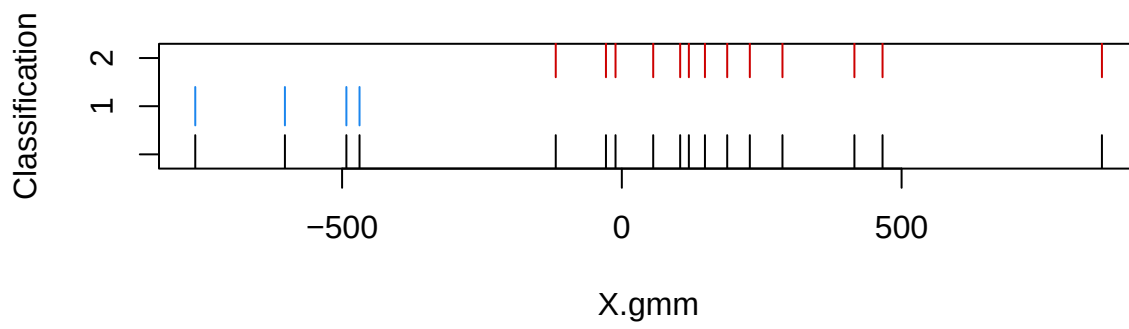


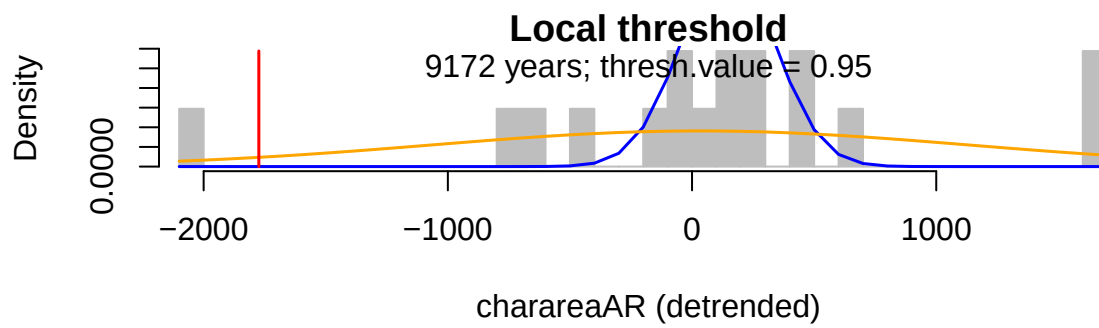
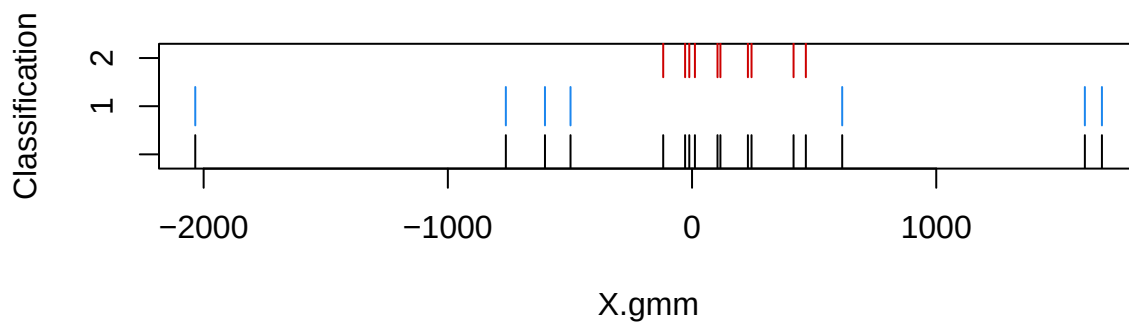


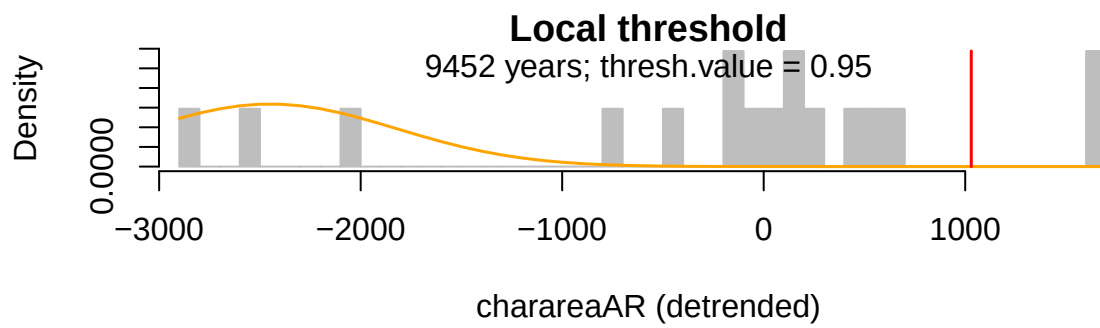
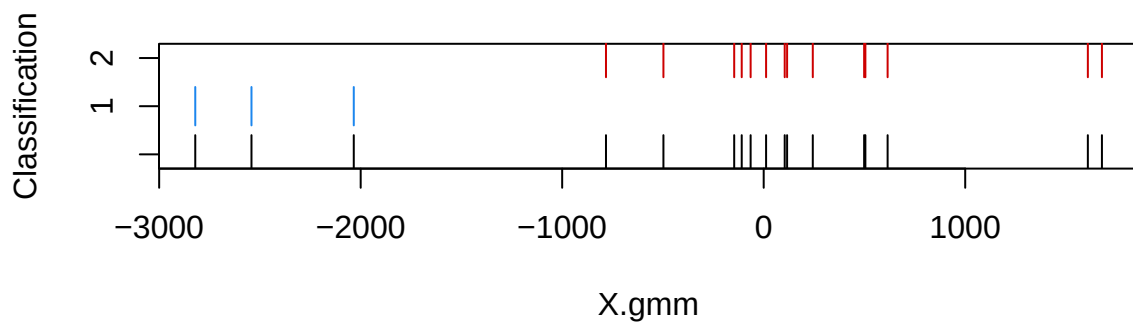


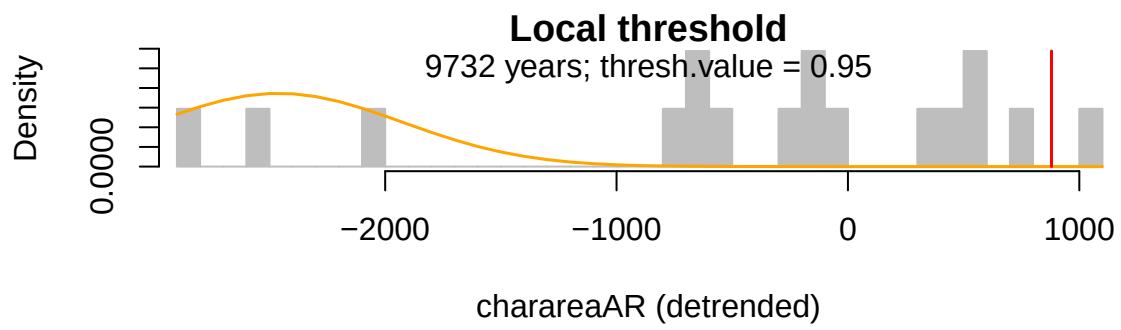
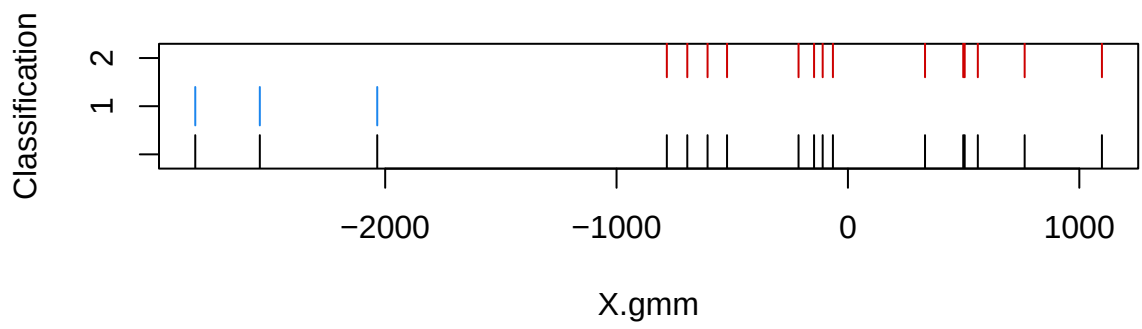


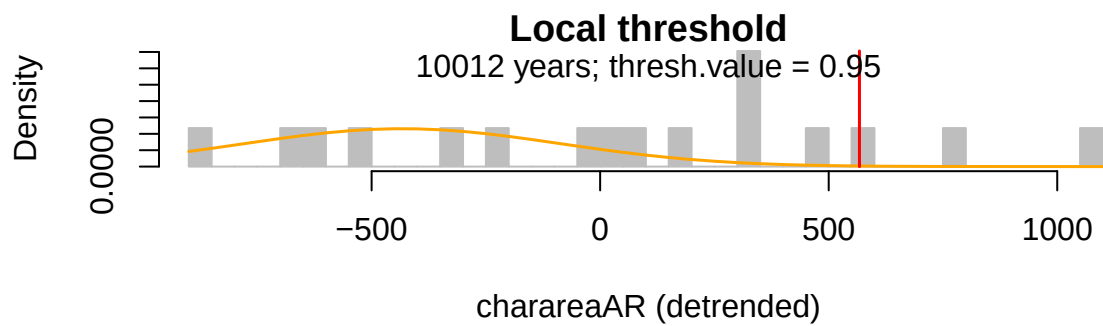
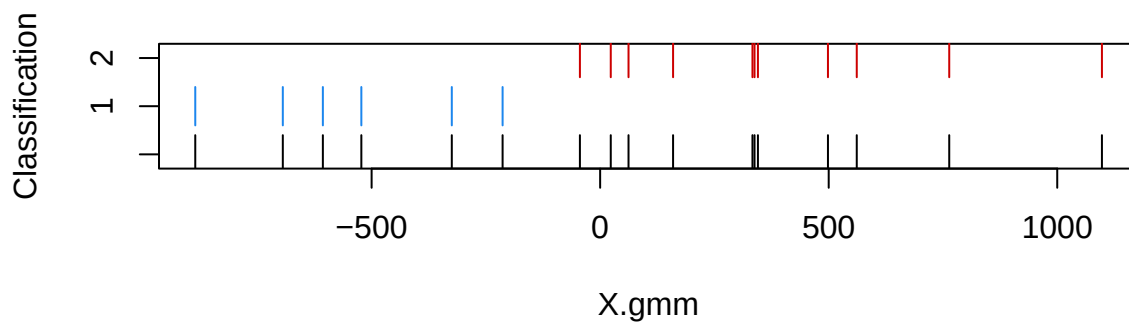


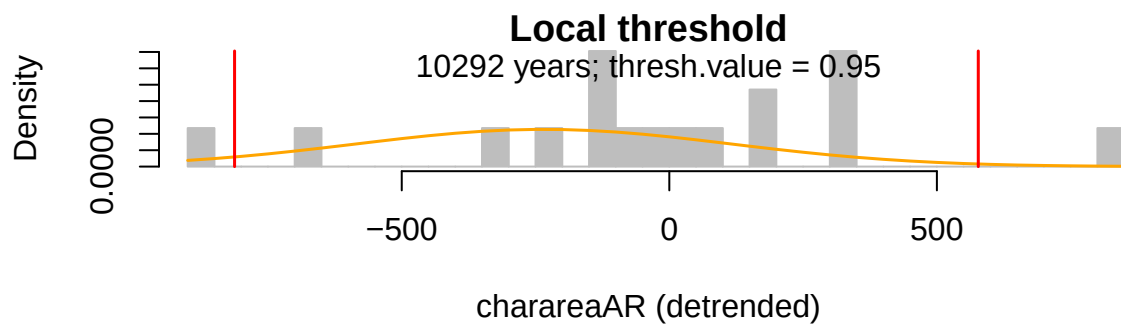
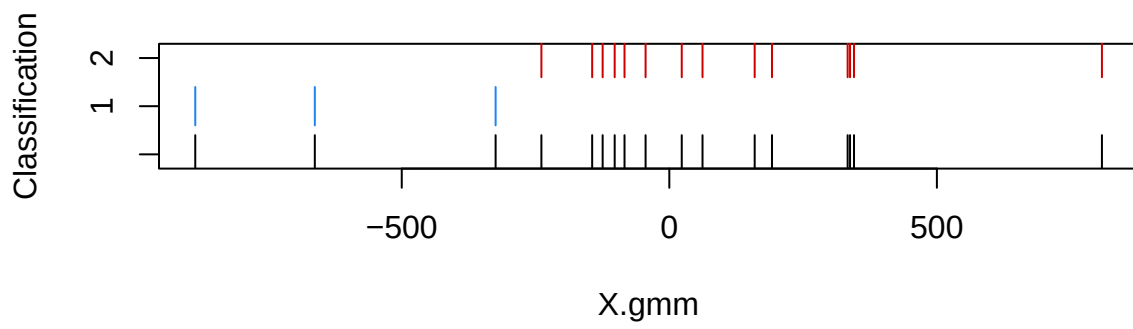


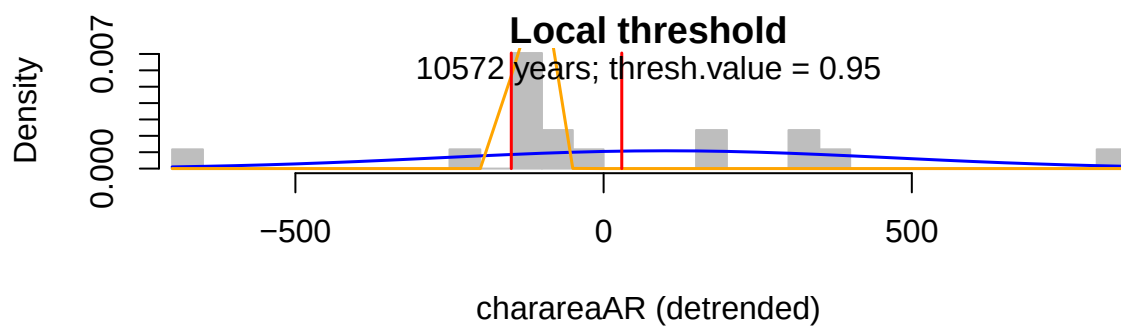
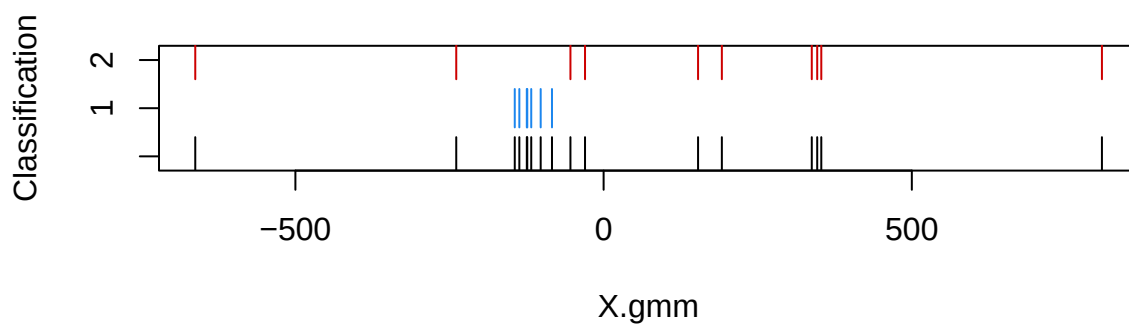


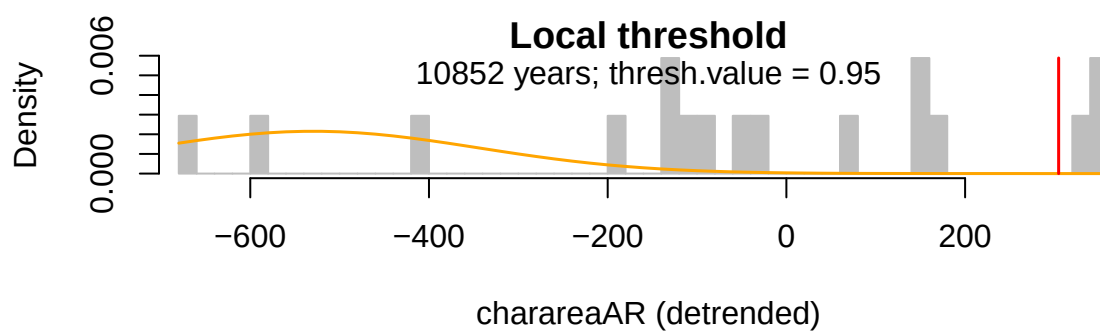
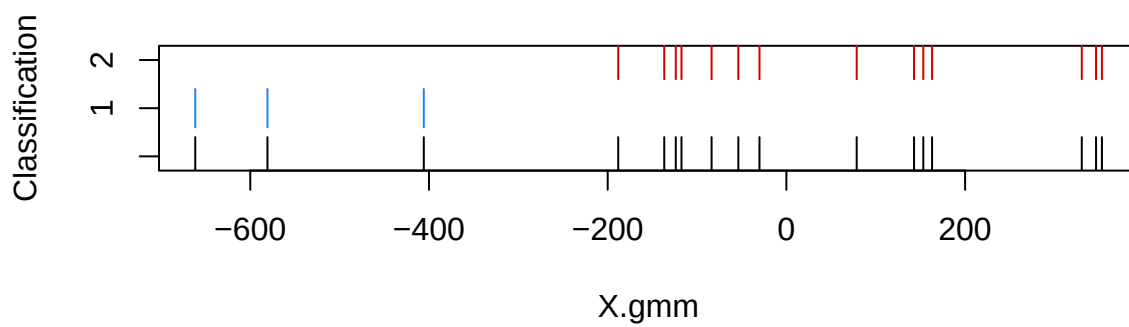


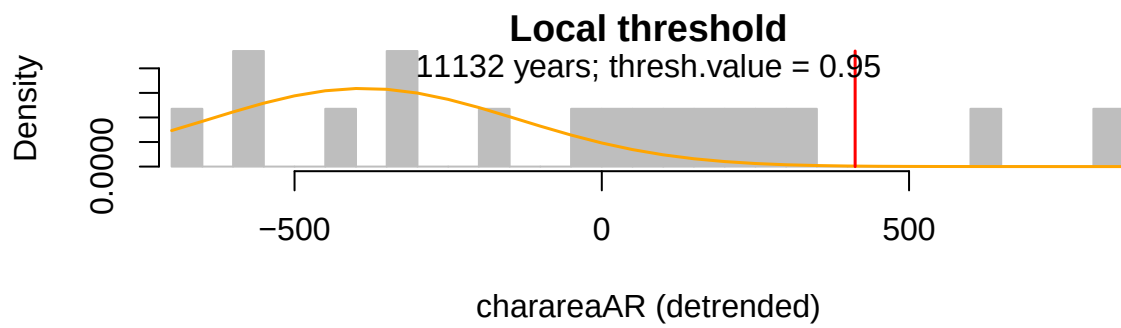
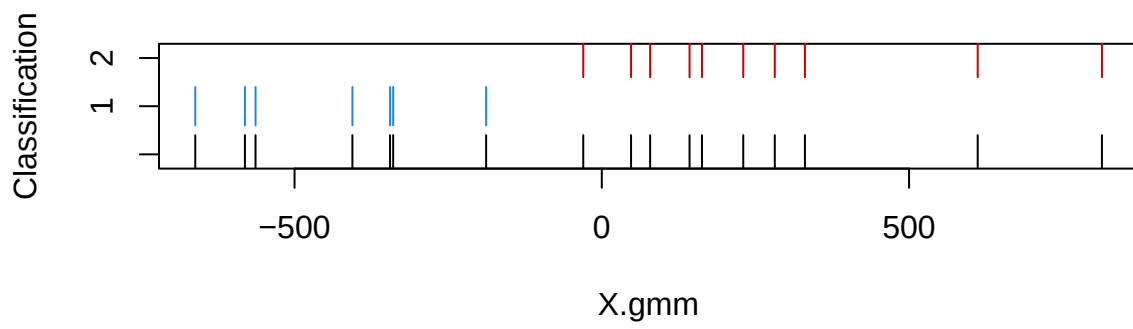


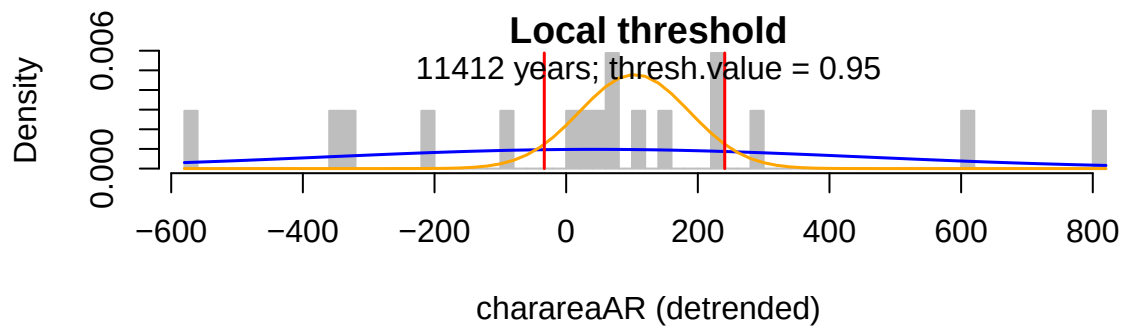
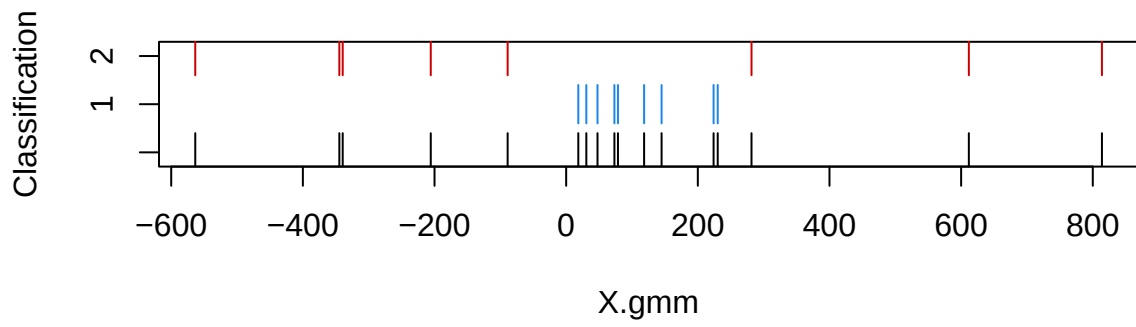


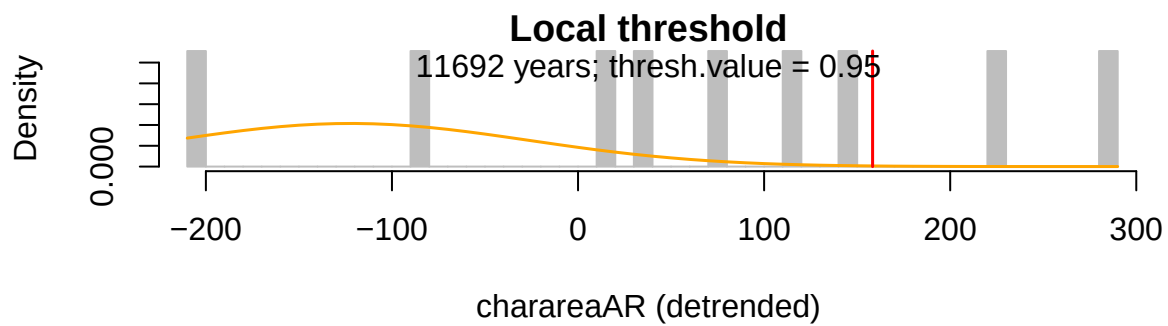
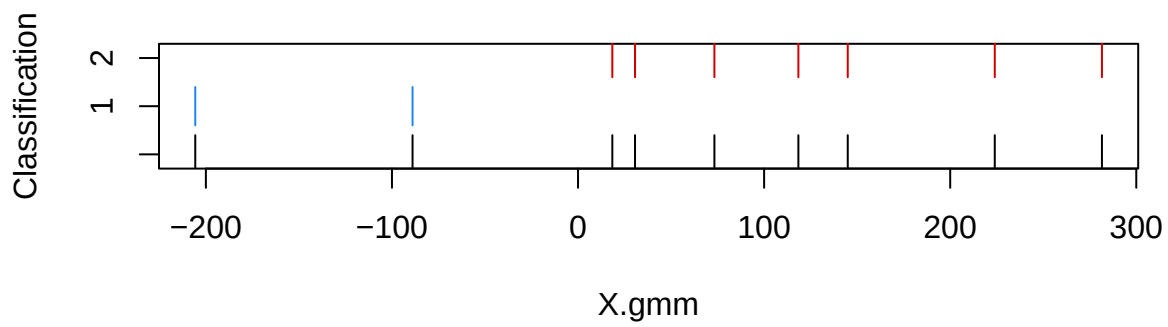


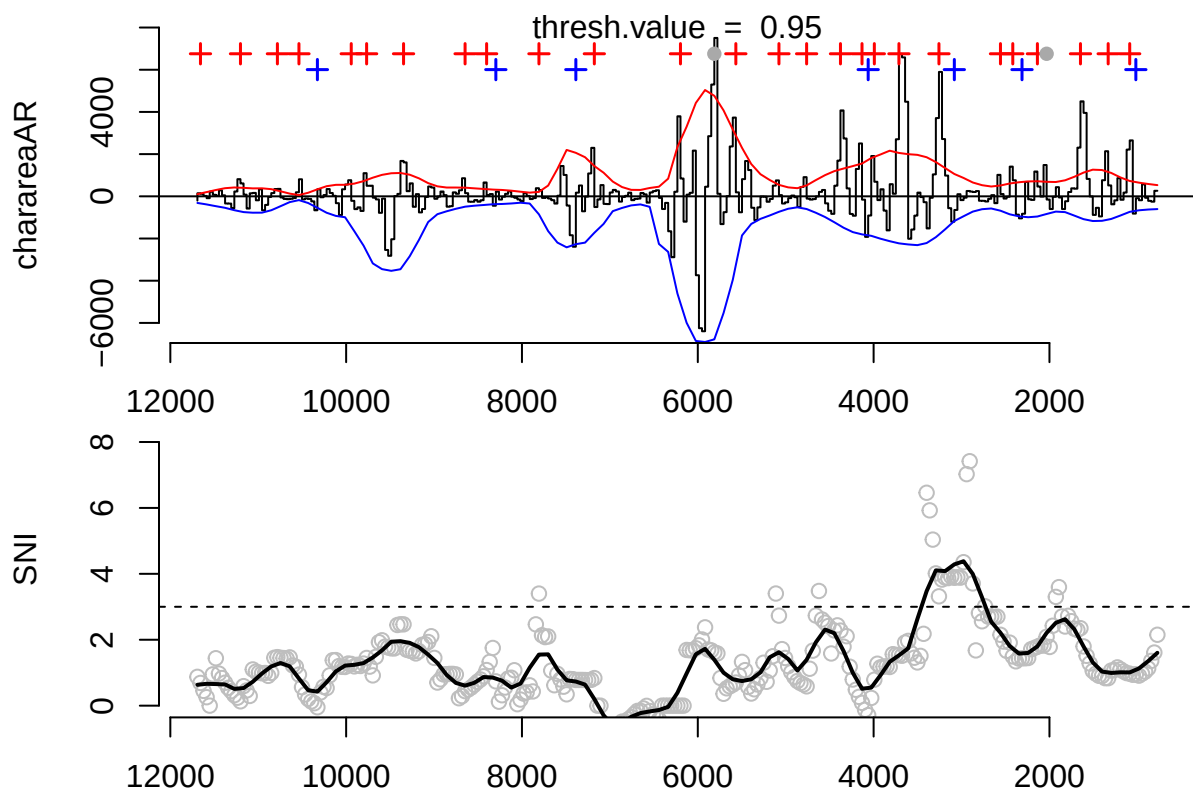




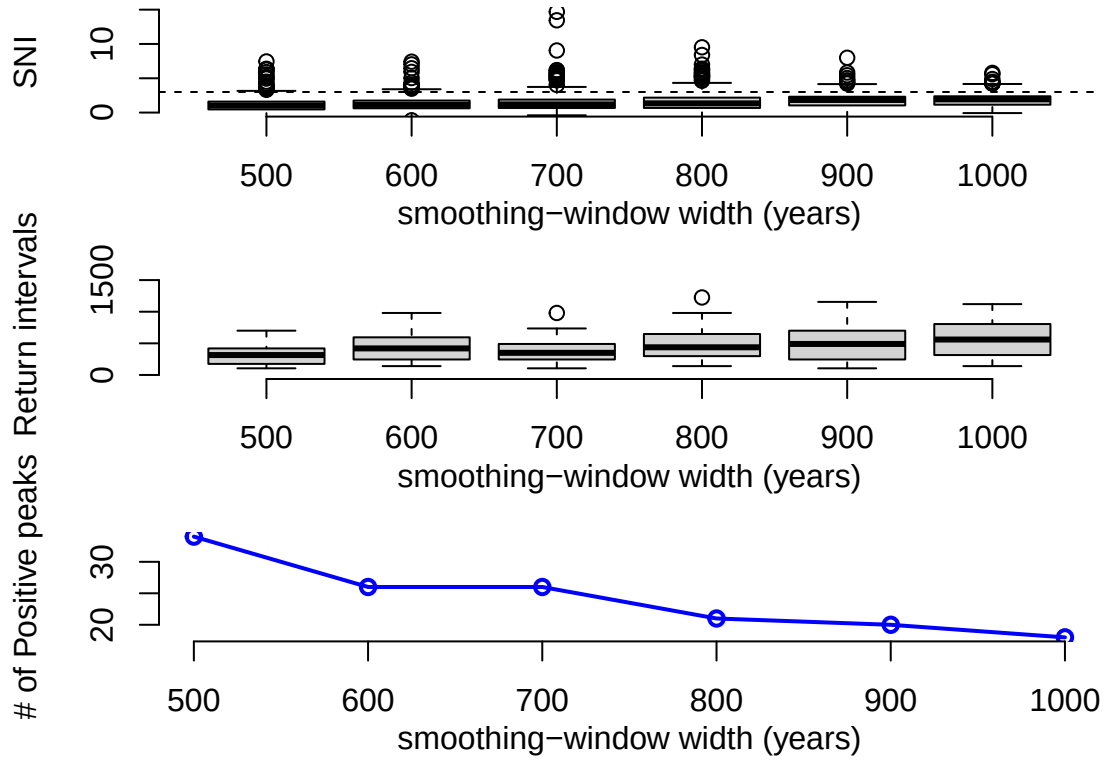


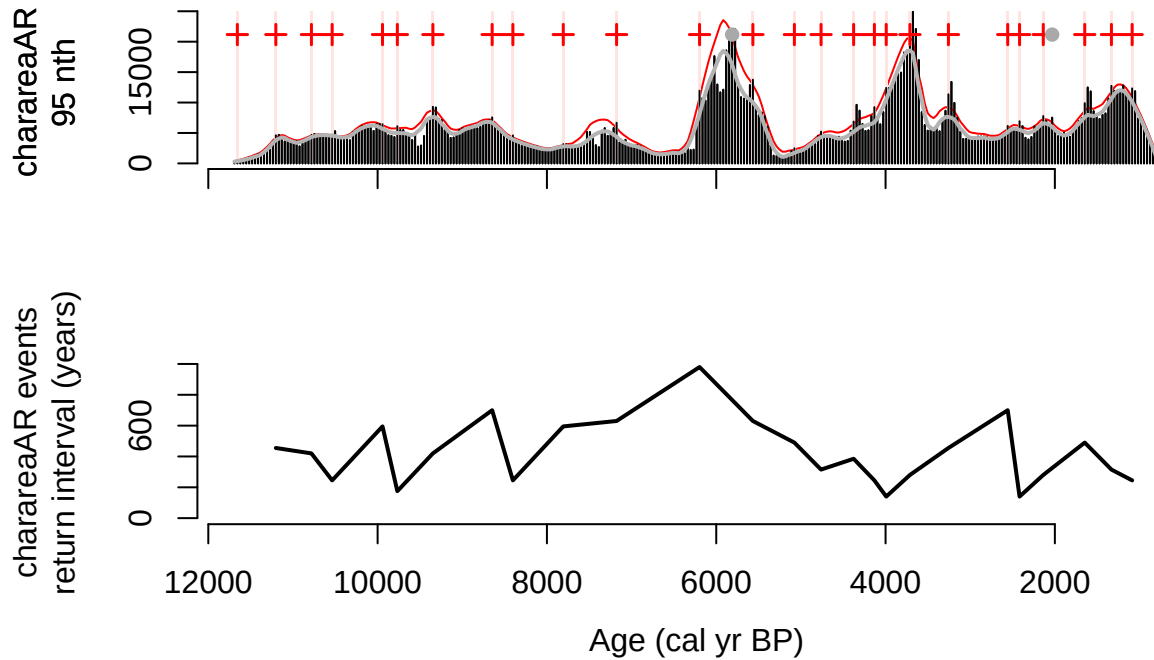






Sensitivity for local threshold at 0.95

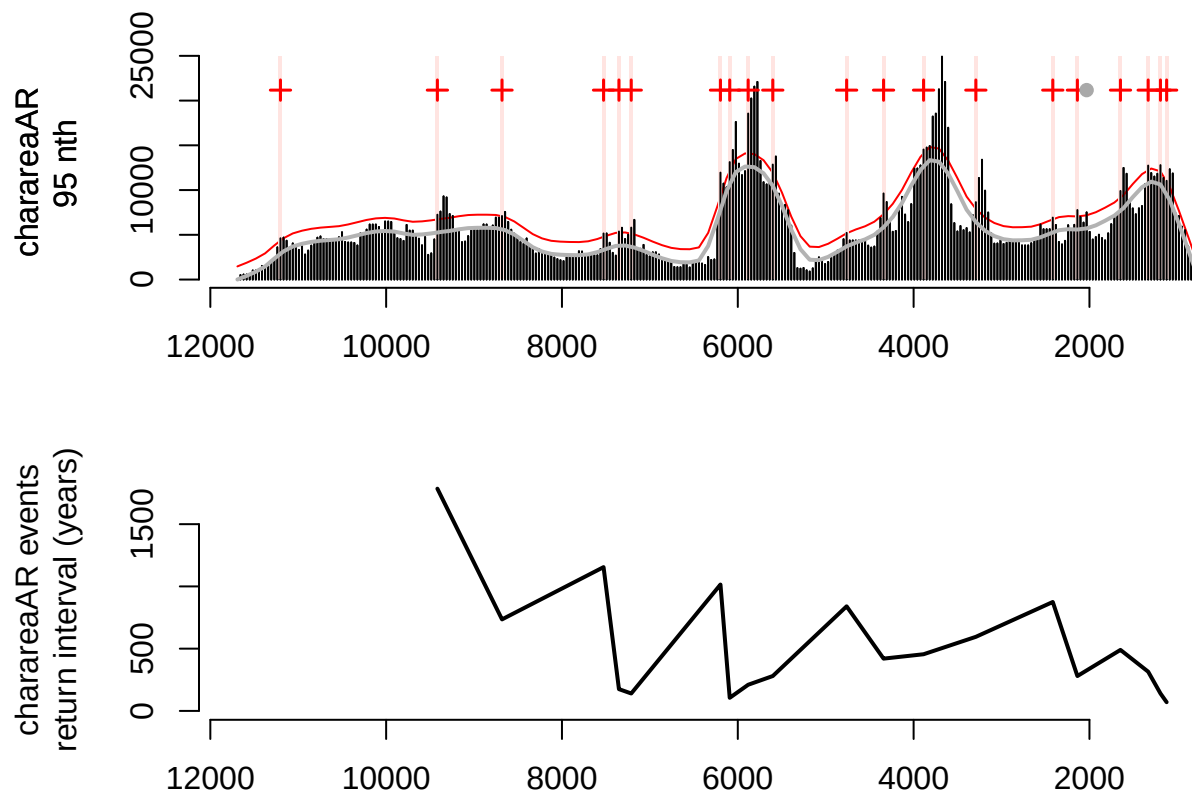




```
tr_loc <- tapas::local_thresh(tr_detr, proxy = "charareaAR",
                              thresh.value = 0.95)
```

When and if you are happy with it, plot results & export data

```
par(mfrow = c(2,1))
tapas::Plot.Anomalies(tr_glob, plot.neg = FALSE, plot.x = TRUE)
tapas::Plot_ReturnIntervals(tr_glob, plot.x = TRUE)
```



```
tr_glob_exp <- tapas::tapas_export(tr_glob)
```

3. CHAR (charcoal particles $\cdot \mu\text{m}^{-2} \cdot \text{yr}^{-1}$)

3.1. Input data

```
head(mass.density) #dataframe with volume and mass information of charcoal samples
```

```
##      SampleId Density..wet.sample...g.cm3. Density..dry.sample...g.cm3.
## 1 TRAM-3U-19-20          1.443714          0.7731429
## 2 TRAM-3U-20-21          1.352385          0.7143846
## 3 TRAM-3U-21-22          1.312714          0.6542143
## 4 TRAM-3U-22-23          1.388818          0.6557727
## 5 TRAM-3U-23-24          1.175850          0.4662500
## 6 TRAM-3U-24-25          1.226727          0.4925000
##      Mass..g.
## 1      1.5025
## 2      1.6349
## 3      1.6743
## 4      1.5234
## 5      1.5846
## 6      1.5901
```

```
head(counting) #dataframe with sampleid and number of charcoal particles found in each sample
```

```
## # A tibble: 6 x 2
##   SampleId      CharCount
##   <chr>          <int>
## 1 TRAM-3U-100-101        3
## 2 TRAM-3U-102-103        3
## 3 TRAM-3U-103-104        1
## 4 TRAM-3U-104-105        1
## 5 TRAM-3U-105-106        3
## 6 TRAM-3U-106-107        1
```

```
head(CharTotS.110.age) #dataframe with sampleid and all charcoal metrics
```

```
##           SampleId CharCount sumCharArea meanCharWL medianCharWL  age depth
## 1 TRAM-3U-100-101        3  47848.073  0.4349000      0.4969 3984  521
## 2 TRAM-3U-102-103        3  45394.326  0.4572667      0.3562 4027  524
## 3 TRAM-3U-103-104        1  13495.610  0.5032000      0.5032 4042  525
## 4 TRAM-3U-104-105        1   8588.116  0.4803000      0.4803 4070  527
## 5 TRAM-3U-105-106        3  28218.094  0.6418333      0.7019 4100  529
## 6 TRAM-3U-106-107        1  15949.358  0.5825000      0.5825 4111  530
```

```
sed.rate.charcoal <- tram20_depth_ages_sed_rate %>% #dataframe with sedimentation rate of charcoal samp
mutate(SampleId = str_replace_all(SampleId, "TRAM20-1B-", "TRAM-")) %>% #change SampleId so it is the
mutate(across(c(depth_diff, age_diff, sed_rate), ~ ifelse(row_number() == 1 & is.na(.x), .x[2], .x)))
filter(SampleId %in% mass.density$SampleId)
```

3.2. Calculate CHAR based on the total area of charcoal particles

```
CHAR <- CharTotS.110.age %>%
  select(SampleId, sumCharArea, age, depth) %>%
  full_join(mass.density, by = "SampleId") %>%
  select(-"Density..wet.sample...g.cm3.") %>%
  rename("Mass" = "Mass..g.",
         "Density" = "Density..dry.sample...g.cm3.") %>%
  left_join(select(sed.rate.charcoal, SampleId, sed_rate), by = "SampleId") %>%
  relocate("age", .after = "SampleId") %>%
  mutate("CHAR.tram" = ((sumCharArea/Mass) * Density * sed_rate)) #μm2·cm-2·yr-1
```

See correlation between CHAR and sedimentation rate, and other metrics

```
cor(CHAR$sed_rate, CHAR$CHAR.tram, method = 'spearman') #cor = 0.28 -> low correlation
```

```
## [1] 0.2858222
```

3.3. Figure CHAR

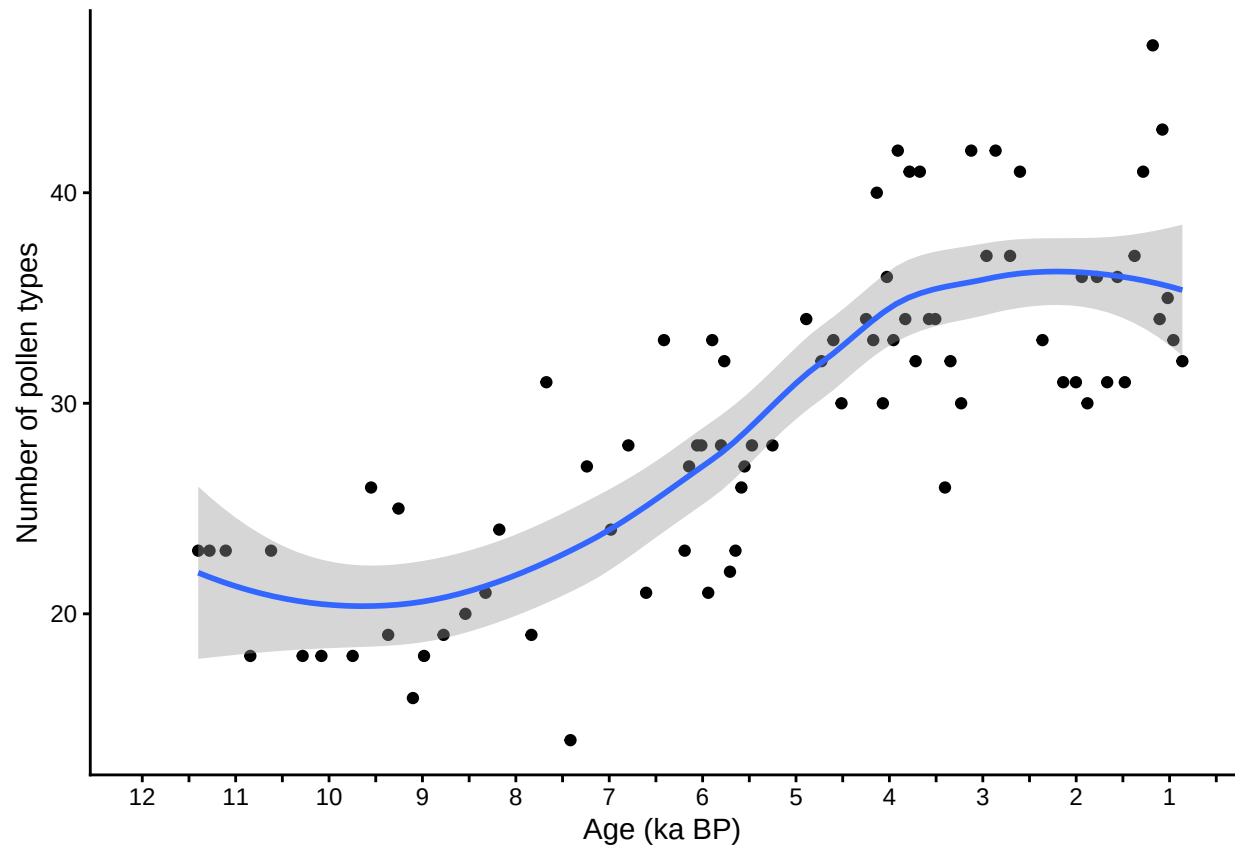

```
pCHAR <- CHAR %>%
  ggplot(aes(x = age/1000, y = CHAR.tram)) +
  geom_col(width = 0.05) +
  ylab("CHAR (charcoal area  $\mu\text{m}^2 \cdot \text{cm}^{-2} \cdot \text{yr}^{-1}$ )") +
  xlab("Age (ka BP)") +
  scale_x_reverse(limits = c(max(CHAR$age)/1000, min(CHAR$age)/1000),
                 breaks = seq(0.5, 12, by = 0.5),
                 labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                "")) +
  theme_classic()
```

4. Tramacastilla pollen

```
head(tram20_pollen_holocene) #Pollen counts throughout the Holocene
```

Plot pollen-type richness

```
tram20_pollen_holocene %>%
  ggplot(aes(x = age/1000, y = `No of taxa`)) +
  geom_point() +
  scale_x_reverse(limits = c(12, 0.863),
                 breaks = seq(0.5, 12, by = 0.5),
                 labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                "")) +
  theme_classic() +
  geom_smooth(method = "loess") +
  labs(x = "Age (ka BP)",
       y = "Number of pollen types")
```



Import pollen groups

```
head(pollen_group)
```

##	Taxon	pollen_group	pollen_group_general	pollen_group_discussion
## 1	Abies	Abies	Abies	Abies
## 2	Pinus	Pinus	Pinus	Pinus
## 3	Juniperus	Other conifer	Juniperus	Juniperus
## 4	Taxus	Other conifer	Evergreen trees	Evergreen trees
## 5	Alnus	Deciduous trees	Other deciduous trees	Other deciduous trees
## 6	Acer	Deciduous trees	Other deciduous trees	Other deciduous trees
##	Functional.group	Community	Functional.fire.response	
## 1	tree	temperate forest	fire-sensitive	
## 2	tree	pinewoods	fire-insensitive	
## 3	tree	shrubland	fire-sensitive	
## 4	tree	temperate forest	fire-sensitive	
## 5	tree	riverine community	fire-sensitive	
## 6	tree	temperate forest	fire-sensitive	
##	Functional.disturbance.response	General.disturbance.sensitivity		
## 1	herbivore-sensitive		high	
## 2	herbivore-insensitive		low	
## 3	herbivore-insensitive		medium	
## 4	herbivore-sensitive		high	
## 5	herbivore-tolerant		medium	
## 6	herbivore-tolerant		medium	
##	Community_2			

```
## 1      Abies_Fagus
## 2      pinewoods
## 3      shrubland
## 4      temperate forest
## 5      riverine community
## 6      temperate forest
```

```
length(unique(pollen_group$pollen_group))
```

```
## [1] 14
```

```
length(unique(pollen_group$pollen_group_general))
```

```
## [1] 37
```

```
length(unique(pollen_group$pollen_group_discussion))
```

```
## [1] 21
```

4.1. Pollen Accumulation Rate (PAR)

Data input

```
head(tram20_density_pollen) #dataframe with density of pollen samples
```

```
## # A tibble: 6 x 2
##   SampleId      Density
##   <chr>      <dbl>
## 1 TRAM20-1B-3U-19-20  0.773
## 2 TRAM20-1B-3U-22-23  0.656
## 3 TRAM20-1B-3U-24-25  0.492
## 4 TRAM20-1B-3U-27-28  0.575
## 5 TRAM20-1B-3U-29-30  0.537
## 6 TRAM20-1B-3U-32-33  0.615
```

```
head(pollen_samples_mass) #dataframe with mass of pollen samples
```

```
##           SampleId Mass
## 1  TRAM20-1B-5U-2-4-5  3.8
## 2  TRAM20-1B-5U-2-14-15  3.8
## 3  TRAM20-1B-5U-2-24-25  3.6
## 4  TRAM20-1B-5U-2-34-35  4.1
## 5  TRAM20-1B-5U-2-44-45  4.5
## 6  TRAM20-1B-5U-2-54-55  3.8
```

```
sed.rate.pollen <- tram20_depth_ages_sed_rate %>% #dataframe with sedimentation rate
  mutate(across(c(depth_diff, age_diff, sed_rate), ~ ifelse(row_number() == 1 & is.na(.x), .x[2], .x)))
  filter(SampleId %in% tram20_pollen_holocene$SampleId)
```

Compute PAR

```

tram20_PAR <- tram20_pollen_holocene %>%
  dplyr::select(~c(`Evergreen trees`:Trees, Shrubs, Herbs, Hygrophytes, Hydrophytes:Spores, `Charcoal <
  left_join(tram20_density_pollen, by = "SampleId") %>%
  left_join(pollen_samples_mass, by = "SampleId") %>%
  left_join(dplyr::select(sed.rate.pollen, SampleId, sed_rate), by = "SampleId") %>%
  mutate(age = as.numeric(age)) %>%
  mutate(across(!c(SampleId, depth, age, `Lycopodium`, Density, Mass, sed_rate),
    ~ (. / ((`Lycopodium` * Mass)/41696)) * Density * sed_rate,
    .names = "PAR_{.col}")) %>% #41696 Lycopodium spores in each sample
  dplyr::select(c(depth, age, PAR_Abies:`PAR_Undetermined`)) %>%
  rename_with(~ gsub("^PAR_", "", .x)) %>% #remove prefix
  dplyr::select(where(~ sum(., na.rm = TRUE) > 0)) #only taxons that appear in these samples

```

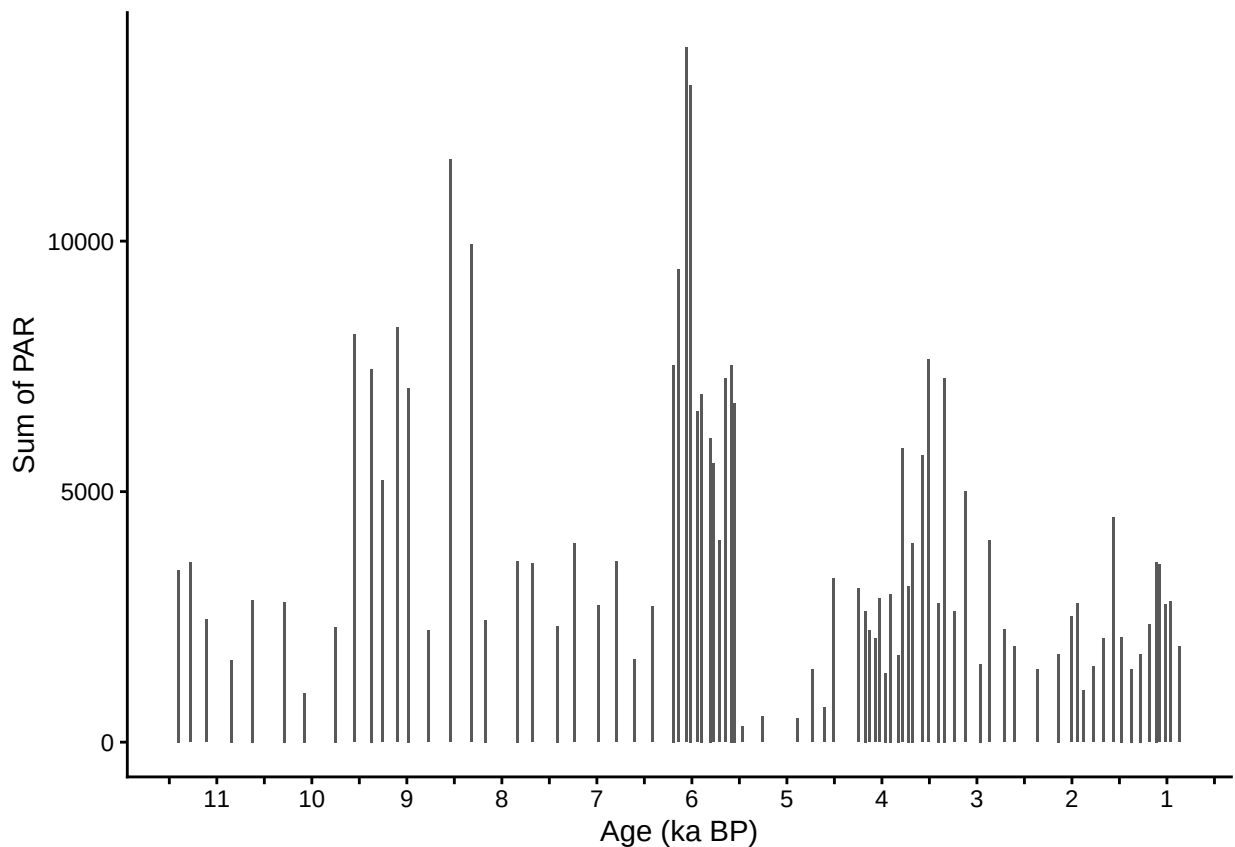
Sum of terrestrial PAR in each sample to see relationship between terrestrial vegetation and CHAR

```

tram20_PAR_sum <- tram20_PAR %>%
  mutate(sum = rowSums(across(Abies:Viola)))

tram20_PAR_sum %>% ggplot(aes(x = age/1000, y=sum)) +
  geom_col() +
  scale_x_reverse(breaks = seq(0.5, 25, by = 0.5),
    labels = ifelse(seq(0.5, 25, by = 0.5) %% 1 == 0, as.character(seq(0.5, 25, by = 0.5)),
  theme_classic() +
  labs(y = "Sum of PAR",
    x = "Age (ka BP)")

```



```

tram20_PAR_CHAR <- tram20_PAR_sum %>%
  left_join(tram20_pollen_holocene %>% select (SampleId, age), by = "age") %>%
  left_join(Char %>% select(age, CHAR.tram), by = "age") %>%
  na.omit()

#check if PAR and CHAR are correlated
cor.test(tram20_PAR_CHAR$sum, tram20_PAR_CHAR$CHAR.tram, method="spearman", exact = FALSE) #34% of the

##
## Spearman's rank correlation rho
##
## data: tram20_PAR_CHAR$sum and tram20_PAR_CHAR$CHAR.tram
## S = 57899, p-value = 0.001545
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.3462138

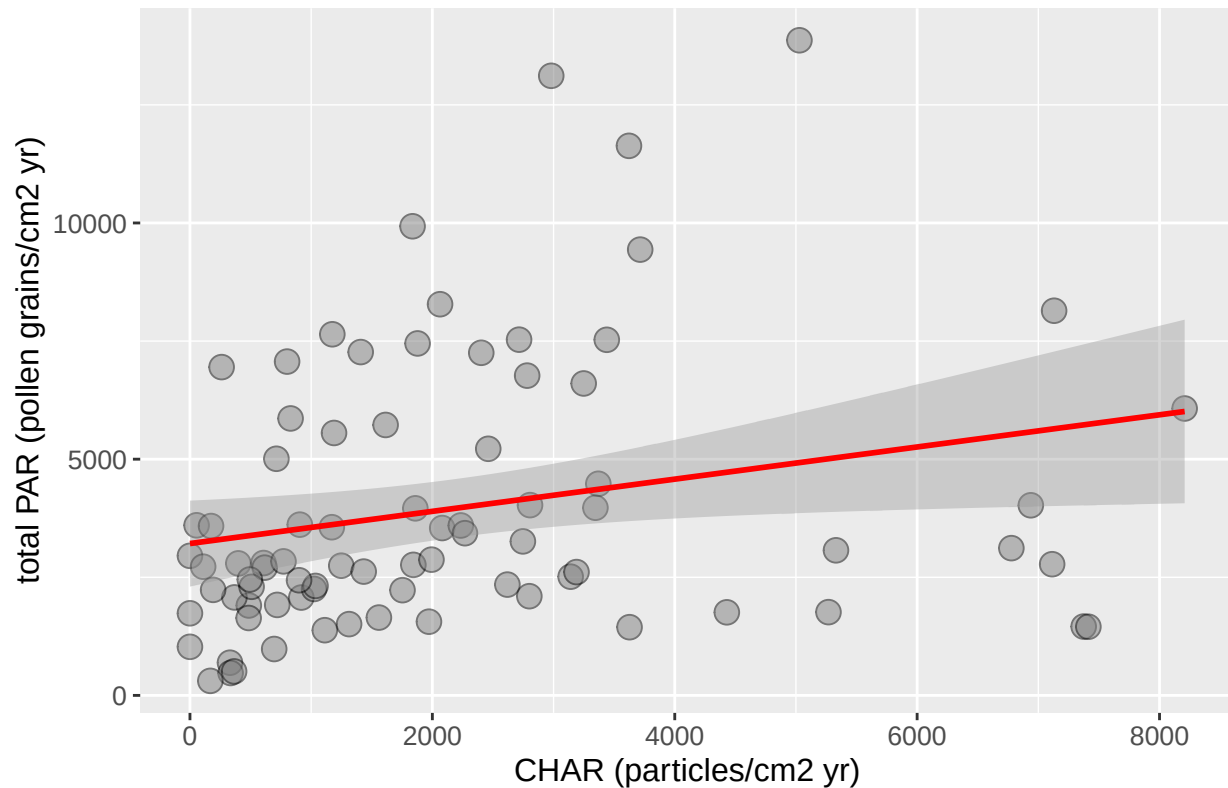
#check if that correlation is linnear or not
summary(lm(tram20_PAR_CHAR$sum~tram20_PAR_CHAR$CHAR.tram)) #6% is the amount of data linnearly correlat

##
## Call:
## lm(formula = tram20_PAR_CHAR$sum ~ tram20_PAR_CHAR$CHAR.tram)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4286.3 -1761.3  -890.4  1551.2  8940.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3214.5628    456.3303   7.044 6.15e-10 ***
## tram20_PAR_CHAR$CHAR.tram    0.3406     0.1535   2.219  0.0293 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2798 on 79 degrees of freedom
## Multiple R-squared:  0.05869,    Adjusted R-squared:  0.04678
## F-statistic: 4.926 on 1 and 79 DF,  p-value: 0.02933

tram20_PAR_CHAR %>%
  ggplot(aes(x = CHAR.tram, y = sum)) +
  geom_point(shape = 21, fill = "gray50", color = "black", size = 4, alpha = 0.5) +
  geom_smooth(method = "lm", col = "red") +
  xlab("CHAR (particles/cm2 yr)") +
  ylab("total PAR (pollen grains/cm2 yr)") +
  ggtitle("PAR vs. CHAR") +
  theme(text = element_text(size = 12),
        plot.title = element_text(size = 16))

```

PAR vs. CHAR



4.1.1. PAR grouped

```
tram20_PAR_groups <- tram20_PAR %>%
  pivot_longer(cols = Abies:Undetermined, names_to = "Taxon", values_to = "PAR_value") %>% #long format
  left_join(pollen_group, by = "Taxon") %>% #Include pollen groups
  group_by(depth, age, pollen_group_general) %>%
  summarise(tram20_PAR_sum = sum(PAR_value, na.rm = TRUE)) %>% #calculate PAR sum for each group
  pivot_wider(names_from = pollen_group_general, values_from = tram20_PAR_sum, values_fill = 0) %>% #ba
  dplyr::select(age, depth, Abies, Pinus, Juniperus, Fagus, Betula, Corylus, `Deciduous Quercus`, Tilia
    `Evergreen trees`, `Other deciduous trees`, Ephedra, Shrubs, Poaceae, `Cerealia type`, Apiaceae
    Cichorioideae, Caryophyllaceae, Chenopodiaceae, Helianthemum, Plantago, Ranunculaceae, Thalictr
    Filipendula, Potentilla, Sanguisorba, Rumex, Scrophulariaceae, Urtica, Urticaceae, `Other herb
    Hygrophytes, Hydrophytes, Undetermined) %>% #keep order
  as.data.frame()
```

4.2. PAR groups for models

```
tram20_PAR_groups_discussion <- tram20_PAR %>%
  pivot_longer(cols = Abies:Undetermined, names_to = "Taxon", values_to = "PAR_value") %>% #long format
  left_join(pollen_group, by = "Taxon") %>% #Include pollen groups
  group_by(depth, age, pollen_group_discussion) %>%
  summarise(PAR_sum = sum(PAR_value, na.rm = TRUE)) %>% #calculate PAR sum for each group
```

```

pivot_wider(names_from = pollen_group_discussion, values_from = PAR_sum, values_fill = 0) %>% #back to
dplyr::select(age, depth, Abies, Pinus, Juniperus, Fagus, Betula, Corylus, `Deciduous Quercus`, Tilia
`Evergreen trees`, `Other deciduous trees`, Shrubs, Poaceae, `Steppe taxa`, `Other herbs`, `An
Hygrophytes, Hydrophytes, Undetermined) %>% #keep order and exclude Ferns and Selaginella
as.data.frame()

tram20_PAR_groups_communities <- tram20_PAR %>%
pivot_longer(cols = Abies:Undetermined, names_to = "Taxon", values_to = "PAR_value") %>% #long format
left_join(pollen_group, by = "Taxon") %>% #Include pollen groups
dplyr::group_by(.data$Community_2, .data$depth, .data$age) %>%
summarise(PAR_sum = sum(PAR_value, na.rm = TRUE)) %>% #calculate PAR sum for each group
filter(Community_2 != "NA") %>% # exclude NAs
pivot_wider(names_from = Community_2, values_from = PAR_sum, values_fill = 0) %>% #back to wide format
as.data.frame()

```

5. Fungal spores

```
head(tram20_spores)
```

```

##           SampleId  age depth Apiosordaria type Chaetomium type
## 1 TRAM20-1B-3U-19-20 863   390              0              0
## 2 TRAM20-1B-3U-22-23 959   395              0              0
## 3 TRAM20-1B-3U-24-25 1020  398              0              0
## 4 TRAM20-1B-3U-27-28 1077  403              0              0
## 5 TRAM20-1B-3U-29-30 1106  406              0              0
## 6 TRAM20-1B-3U-32-33 1180  411              1              0
##   Coniochaeta cf. lignaria Delitschia type Hypocopra type Podospora type
## 1                      0              0              0              3
## 2                      0              0              0              2
## 3                      0              0              0              1
## 4                      0              0              0              0
## 5                      1              0              0              0
## 6                      0              0              0              1
##   Arnium imitans Arnium type Cercophora type Schizothecium conicum
## 1              2              1              0              0
## 2              0              1              2              0
## 3              1              0              1              0
## 4              1              0              3              0
## 5              0              0              0              0
## 6              0              0              1              0
##   Schizothecium inaequale Sphaerodes fimicola Gelasinospora type
## 1                      0              0              0
## 2                      0              0              0
## 3                      0              0              0
## 4                      0              0              0
## 5                      0              0              0
## 6                      0              0              0
##   Neurospora type Sordaria type (HRV-55A) Sporormiella type
## 1              0              9              20
## 2              0              4              6

```

## 3	0	4	4
## 4	0	5	5
## 5	0	2	15
## 6	0	2	2
##	Trichodelitschia type	Other Sordariales	Alternaria Cirrenalia
## 1	0	1	0
## 2	0	0	1
## 3	0	0	2
## 4	1	0	1
## 5	0	0	5
## 6	0	0	2
##	Diporotheca webbiae	Kretzschmaria deusta	Puccinia Tilletia
## 1	0	2	0
## 2	0	1	0
## 3	0	0	0
## 4	0	1	0
## 5	1	2	0
## 6	1	0	0
##	Valsaria variospora	Glomus Lycopodium clavatum	Obligate coprophilous
## 1	0	3	93
## 2	0	3	59
## 3	0	1	40
## 4	0	1	59
## 5	0	2	75
## 6	0	1	82
##	Facultative coprophilous	total_spores	Noftaxa
## 1	4	41	8
## 2	3	20	8
## 3	2	14	7
## 4	4	18	8
## 5	1	28	7
## 6	2	11	8

5.1. Compute Spore Accumulation Rate (SAR)

```
tram20_SAR <- tram20_spores %>%
  left_join(tram20_density_pollen, by = "SampleId") %>%
  left_join(pollen_samples_mass, by = "SampleId") %>%
  left_join(dplyr::select(sed.rate.pollen, SampleId, sed_rate), by = "SampleId") %>%
  mutate(age = as.numeric(age)) %>%
  mutate_all(~replace(., is.na(.), 0)) %>%
  mutate(across(!c(SampleId, depth, age, `Lycopodium clavatum`, Density, Mass, sed_rate),
    ~ (. / ((`Lycopodium clavatum` * Mass)/41696)) * Density * sed_rate,
    .names = "PAR_{.col}")) %>% #41696 lycopodium spores in each sample
  dplyr::select(c(depth, age, `PAR_Apiosordaria type`:`PAR_Facultative coprophilous`)) %>%
  rename_with(~ gsub("^PAR_", "", .x)) %>% #remove prefix
  mutate(total_SAR = rowSums(across(`Apiosordaria type`:Glomus)))
```

Import spore groups


```
head(spores_groups)
```

```
## # A tibble: 6 x 2
##   Taxon                'Spore type'
##   <chr>                <chr>
## 1 Apiosordaria type    Facultative coprophilous
## 2 Chaetomium type      Facultative coprophilous
## 3 Coniochaeta cf. lignaria Facultative coprophilous
## 4 Delitschia type      Obligate coprophilous
## 5 Hypocopra type       Facultative coprophilous
## 6 Podospora type       Facultative coprophilous
```

Obtain final data set

```
tram20_SAR_long <- tram20_SAR %>%
  pivot_longer(cols = `Apiosordaria type`:total_SAR,
    names_to = "Taxon", values_to = "SAR") %>%
  left_join(spores_groups, by = "Taxon") %>%
  arrange(age, `Spore type`)

tram20_SAR_coprophilous <- tram20_SAR_long %>%
  filter(`Spore type` == "Facultative coprophilous" |
    `Spore type` == "Obligate coprophilous") %>%
  dplyr::select(-`Spore type`) %>%
  pivot_wider(names_from = Taxon, values_from = SAR)
```